Plataformas de Desenvolvimento 2024/2025

Mestrado em Informática e Sistemas

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Outline

What's this course section about?

"Development Platforms"

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Objectives

- The main objective of this course is to provide knowledge, understanding and practice on cloud
  - Define Cloud Concepts
  - Identify the global infrastructure components of AWS
  - Explain AWS Cloud architectural principles

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Course Philosophy

- Theoretical classes with presentation of the subjects and study materials.
- Out of the class work
  - Study
  - Assignments for feedback
  - Graded assignments
- Labs with
  - Practical work

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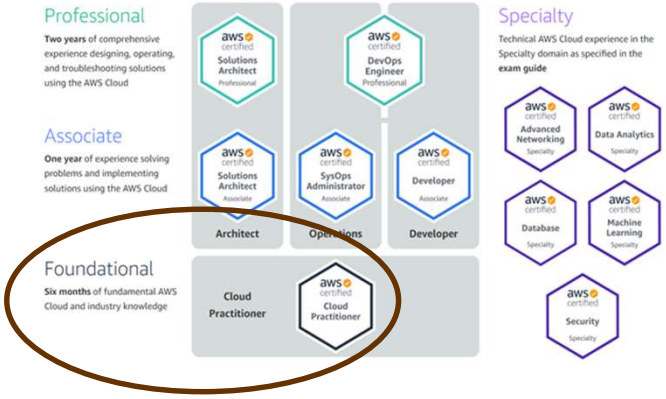
Contents

1. Cloud Computing Overview
2. Cloud Concepts Overvi
3. Cloud Economics and Billing
4. AWS Global Infrastructure Overview and Architecture
5. AWS Cloud Security
6. AWS Computational Resources
7. Databases
8. AWS IoT and AWS SNS
9. API Gateway and REST
10. Auto Scaling and Monitoring

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Contents

Course Section Based on Content for AWS Foundational Certification



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Matching between AWS course and PD

AWS Cloud Foundation	To be evaluated?
Module 1	Y
Module 2	Y
Module 3	Y
Module 4	Y
Module 5	Partially (VPC)
Module 6	Partially (EC2 and Lambda)
Module 7	N
Module 8	Partially (RDS and DynamoDB)
Module 9	Partially
Module 10	N

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Bibliography (always in progress...)

- Mastering Cloud Computing - *Foundations and Applications Programming* Authors: Rajkumar Buyya, Christian Vecchiola and S. Thamarai Selvi
- AWS Academy
- Several papers.

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Support

- All the material available at Microsoft Teams (https://teams.microsoft.com/)
  - Also **inforestudante.ipc.pt** when necessary (e.g. summaries)
- Communication between instructor and students preferably using General chat or direct contacts at MS Teams
- Assignments delivered at MS Teams
- Grades and feedback to students through MS Teams

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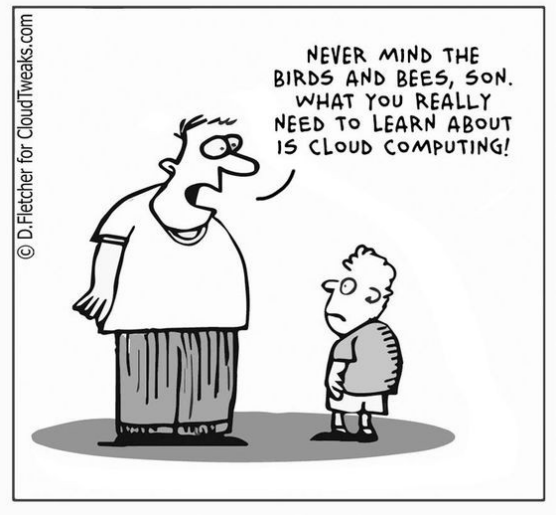
2. Cloud Computing Overview

what is cloud computing?

Ricardo Filipe, ISEC (ricardo.filipe@isec.pt)

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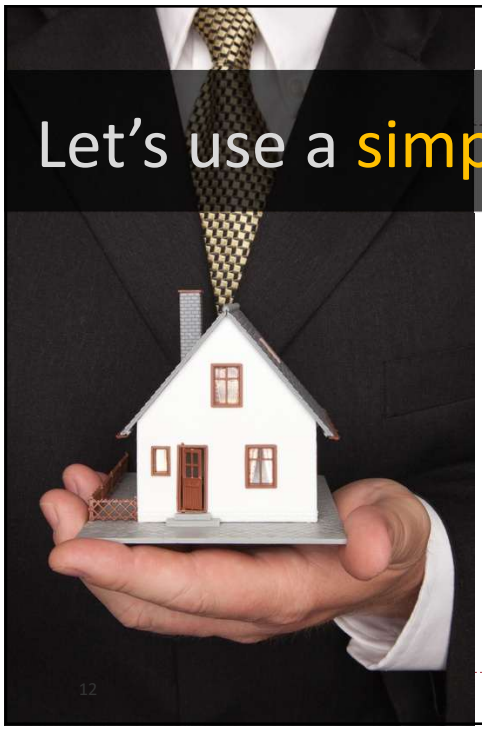
What is Cloud Computing?



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Let's use a simple analogy



Say you just moved to a city, and you're looking for

a place to live

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What is Cloud Computing?

- You can either **Build** a house or **Rent** an apartment

- If you **build** a house, there are a few important **decisions** you have to make:
  - How big is the house?
  - Are you planning to grow a large family?
  - Customize the house:
    - Gardens, colors, furnitures…
  - Do you want to remodel in the future after the house is built?



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Once the house is built, you're responsible for maintenance



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How about renting?

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What is Cloud Computing?

- Consider a **builder** in your city builds a massive number of apartment units
- You make a fewer, simpler decisions
- You can start with one unit and grow later, or downsize
- No need to worry about maintenance cost
- No need to Hire landscapers, electricians, plumbers
- But: You **do not** have a lot of options to customize your unit



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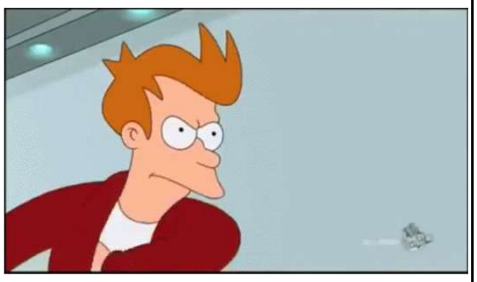
What is Cloud Computing?

Pay as You Go

Just pay your rent

and utilities





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What's the analogy with Cloud?

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What is Cloud Computing?

- Buying a house is like owning a datacenter
  - On-premise software
- Renting an apartment is like Computing as a service. More precisily:
  - Software as a Service (SaaS)
  - Platform as a Service (PaaS)
  - Infrastructure as a Service (IaaS)
- Sharing and virtualizing resources
- Outsourcing infrastructures
- Reduce maintenance
- Reduce costs
  - Why pay for more computing capacity than you need?

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Cloud Computing – history

- Cloud Computing: a *new* technology?
  - Presented in 1966: "***The Challenge of the Computer Utility" by Douglas F. Parkhill***

"The computer utility is a service organization which makes available some or all of the facilities of a large-scale electronic computer to the public on a pay-as-you-go basis. The customer rents from the computer utility time on the electronic computer, a rental which includes both the use of the central processor and the use of input and output devices."

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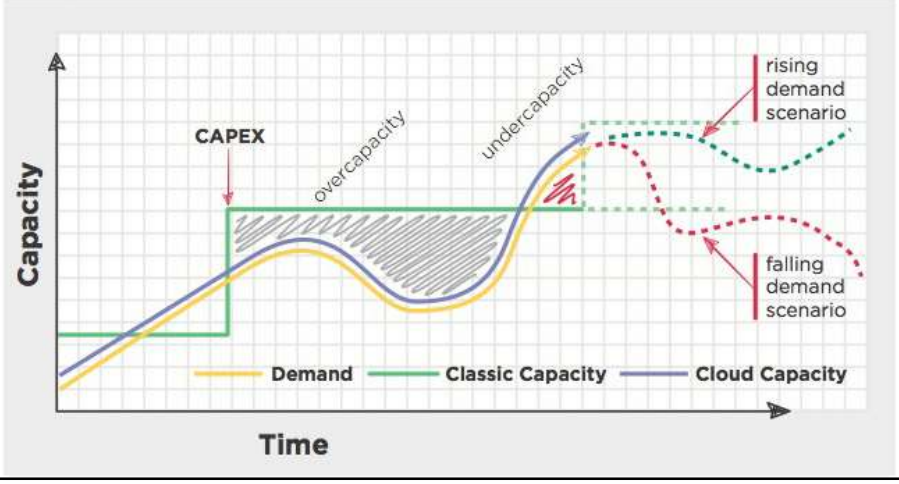
Cloud Computing – timeline

- **Introduction to computers**, provided for timesharing by IBM and DEC, in the 1960s
- **Emergence of the first VM**, developed by IBM, 1972
- First **use of Cloud as a symbol**, 1977
- **Advent of internet** and **world wide web**, 1991
- Telecommunication companies started offering **VPN and other online services**
- The word ***cloud*** **was first tossed by Ramesh Chellappa**, 1997
- Incorporation of **Salesforce and VMware**, 1999
- Launch of **Amazon Web Services** by Amazon, 2002
- Launch of **App Engine by Google**, 2008
- Launch of **Microsoft Azure by Microsoft Corporation**, 2010

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Cloud Computing advantages from client point-of-view

- Less upfront Capex
- Don't need to provide resources for peak periods
- Pay-as-you go architecture
- Focus in core business
- Lower costs?



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Cloud Computing

advantages from cloud provider point-of-view

- **Hardware re-utilization:** many companies only need the computational power in peak hour; to recover the initial investment, companies rent their facilities during off-peak hours.
- **Licensing costs, energy used and staff wages:** large data centers can obtain software licensing with lower prices; additionally, since electricity prices and wages depend on the region, cloud providers can relocate the data center to minimize these expenses.
- E.g. Cost difference between medium and large data center:

	Medium-sized DC	Very Large DC	Ratio (Large-to-Small DC)
Network	$95 per Mbit/sec/month	$13 per Mbit/sec/month	7.1
Storage	$2.20 per GByte/month	$0.40 per GByte/month	5.7
Administration	140 Servers/ Admin	>1000 Servers/ Admin	7.1

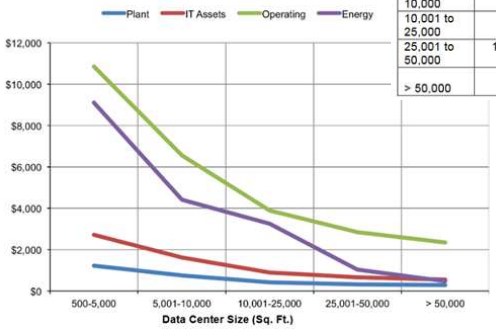
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Cloud Computing

advantages from cloud provider point-of-view

	Medium-sized DC	Very Large DC	Ratio (Large-to-Small DC)
Network	$95 per Mbit/sec/month	$13 per Mbit/sec/month	7.1
Storage	$2.20 per GByte/month	$0.40 per GByte/month	5.7
Administration	140 Servers/ Admin	>1000 Servers/ Admin	7.1

Data Center Size (sq. ft.)	No. of Racks	Avg Compute Load (kW)	Avg Rack Density (kW)	Average Cost Per kW		Average Cost Per Rack	
				Annually	Monthly	Annually	Monthly
500 to 5,000	28.5	105	3.5	$26,495	$2,208	$97,614	$8,134
5,001 to 10,000	58	318	5.7	$13,662	$1,135	$74,689	$6,224
10,001 to 25,000	95	620	6.5	$8,464	$705	$55,242	$4,604
25,001 to 50,000	128.5	972	8	$6,734	$561	$50,841	$4,245
> 50,000	183	1,400	7.8	$5,467	$456	$41,825	$3,485



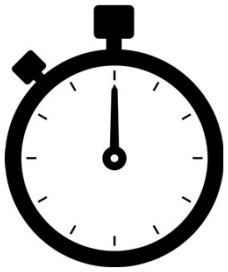
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Cloud Computing advantages referenced by AWS

Advantage #1—Trade capital expense for variable expense:



Data center investment based on forecast



Pay only for the amoun you consume

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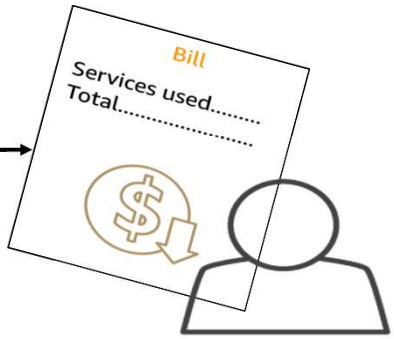
Cloud Computing advantages referenced by AWS

Advantage #2—Benefit from massive economies of scale:

AWS Cloud

Economies of scale

Savings



Because of aggregate usage from all customers, AWS can achieve higher economies of scale and pass savings on to customers.

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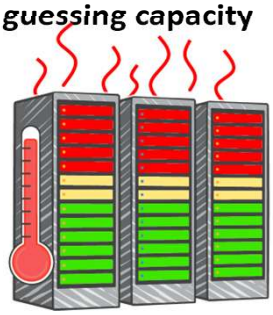
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Cloud Computing advantages referenced by AWS

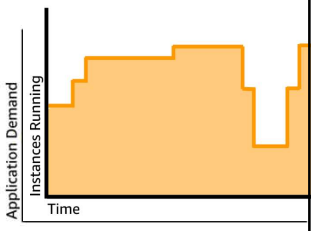
Advantage #3—Stop *guessing* capacity



Overestimated server capacity



Underestimated server capacity



Scaling on demand

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Cloud Computing advantages referenced by AWS

Advantage #4—Increase speed and agility

Weeks between wanting resources and having resources





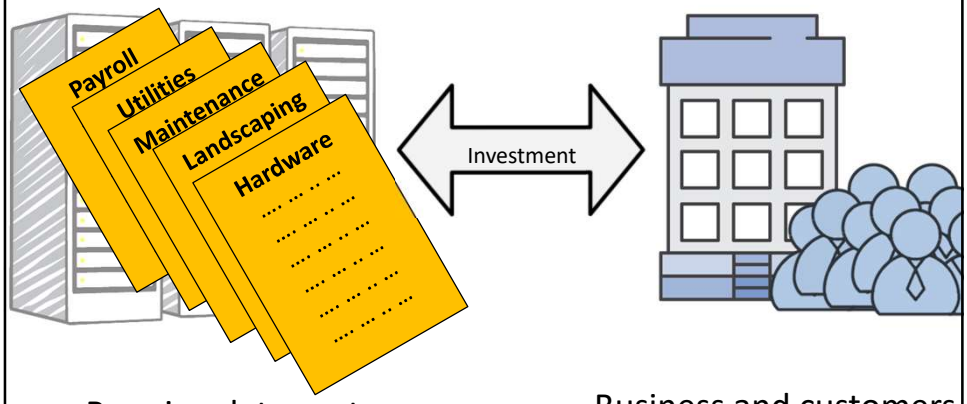
Minutes between wanting resources and having resources

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Cloud Computing
advantages referenced by AWS

Advantage #5—Stop spending money on running and maintaining data

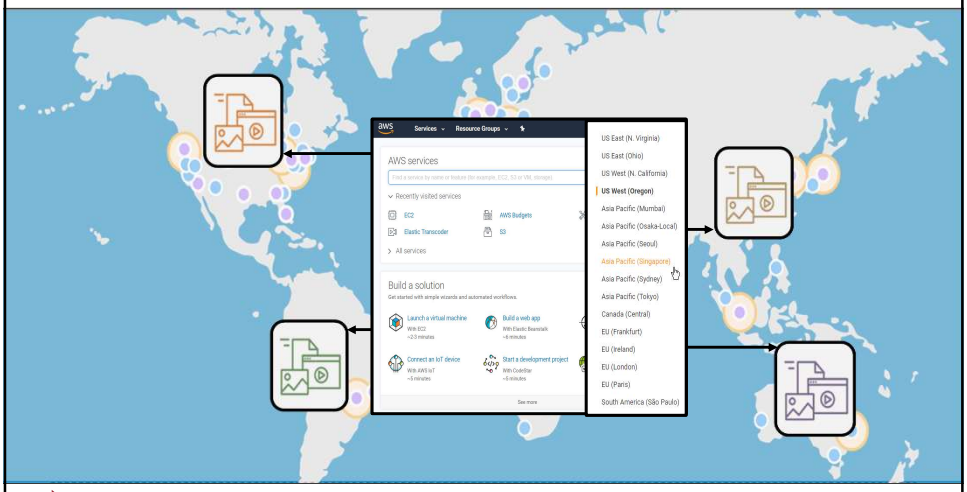


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Cloud Computing
advantages referenced by AWS

Advantage #6—Go global in minutes



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Cloud Computing – NIST Definition

"Cloud computing is a model for enabling **ubiquitous, convenient, on-demand network access** to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. This cloud model is composed of **five** essential characteristics**, three** service models, and **four** deployment models."

Also know as NIST 5–3–4 Rule or Principle of Cloud Computing

NIST - National Institute of Standards and Technology

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Cloud Computing
NIST 5–3–4 Rule/Principle of Cloud Computing

- NIST 5-3-4 Rule/Principle of Cloud Computing offers a structured framework for organizations to assess and adopt cloud computing models and services.

5 Characteristics	3 Service Models	4 Deployment models
1. On-demand self-service	1. SaaS: Software as a service	1. Private cloud
2. Broad network access	2. PaaS: Platform as a service	2. Public cloud
3. Resource pooling	3. IaaS: Infrastructure as a service	3. Hybrid cloud
4. Rapid elasticity		4. Community cloud
5. Measured service		

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Cloud Computing
NIST 5–3–4 Rule/Principle of Cloud Computing

- 5 Characteristics:
  - ***On-demand self-service.*** A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.
  - ***Broad network access.*** Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).

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Cloud Computing
NIST 5–3–4 Rule/Principle of Cloud Computing

- 5 Characteristics:
  - ***Resource pooling.*** The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand. There is a sense of location independence in that the customer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter). Examples of resources include storage, processing, memory, and network bandwidth.

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Cloud Computing
NIST 5–3–4 Rule/Principle of Cloud Computing

- 5 Characteristics:
  - ***Rapid elasticity.*** Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time.
  - ***Measured service.*** Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

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Cloud Computing
NIST 5–3–4 Rule/Principle of Cloud Computing

- 3 service Models:
  - ***Software as a Service (SaaS).*** The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure. The applications are accessible from various client devices through either a thin client interface, such as a web browser (e.g., web-based email), or a program interface. **The consumer does not manage or control the underlying cloud infrastructure** including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings.

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Cloud Computing

NIST 5–3–4 Rule/Principle of Cloud Computing

- 3 service Models:
  - ***Platform as a Service (PaaS).*** The capability provided to the consumer is **to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages, libraries, services, and tools supported by the provider.** The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly configuration settings for the application-hosting environment.

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Cloud Computing

NIST 5–3–4 Rule/Principle of Cloud Computing

- 3 service Models:
  - ***Infrastructure as a Service (IaaS).*** The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications. The consumer does not manage or control the underlying cloud infrastructure but **has control over operating systems, storage, and deployed applications**; and possibly limited control of select networking components (e.g., host firewalls).

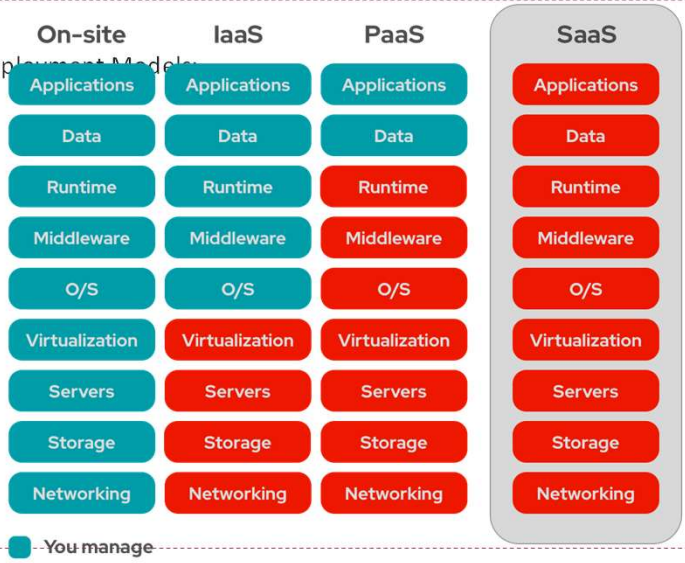
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Cloud Computing

NIST 5–3–4 Rule/Principle of Cloud Computing

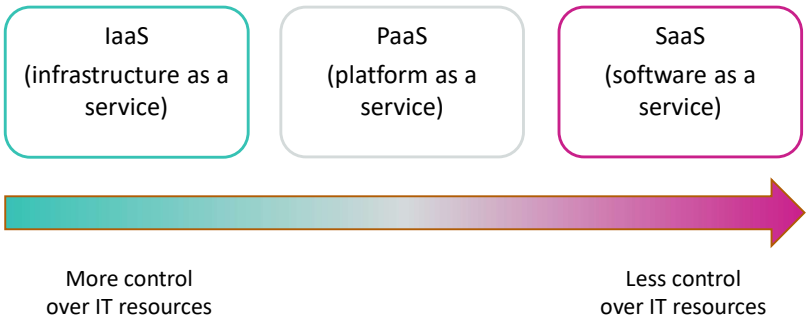
- 3 Deployment Models:



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Cloud service models



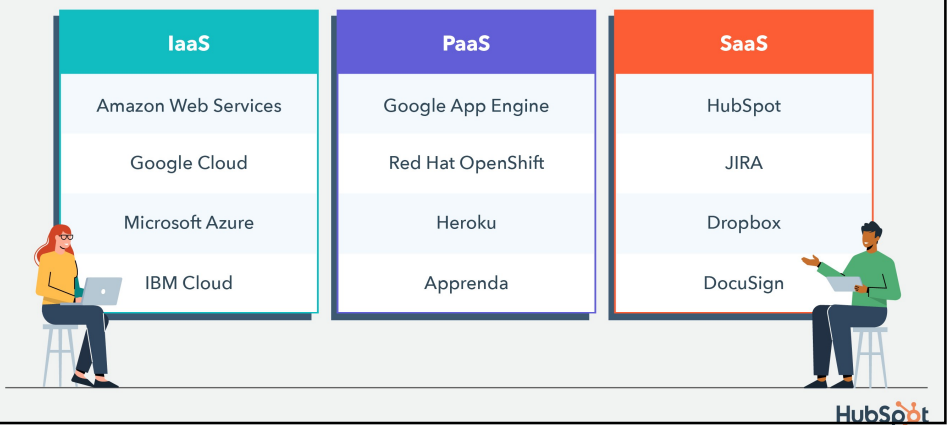
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Cloud Computing

NIST 5–3–4 Rule/Principle of Cloud Computing

IaaS vs. PaaS vs. SaaS Examples



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Cloud Computing

NIST 5–3–4 Rule/Principle of Cloud Computing

4 deployment models:

- *Private cloud.* The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units). It may be **owned, managed, and operated by the organization**, a third party, or some combination of them, and it may exist on or off premises.

- *Community cloud.* The cloud infrastructure is provisioned for exclusive use by a specific **community of consumers** from organizations that have shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.

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Cloud Computing

NIST 5–3–4 Rule/Principle of Cloud Computing

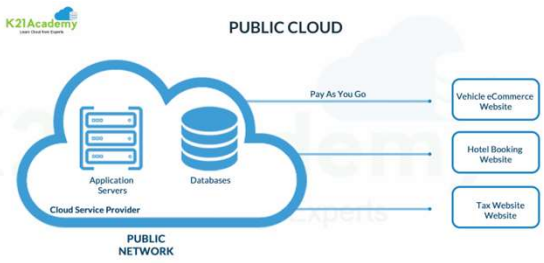
4 deployment models:

- *Public cloud.* The cloud infrastructure is provisioned for **open use by the general public.** It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider.
- *Hybrid cloud*. The cloud infrastructure is a **composition of two or more distinct cloud infrastructures** (private, community, or public) that remain unique entities, but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds).

Cloud Computing

NIST 5–3–4 Rule/Principle of Cloud Computing

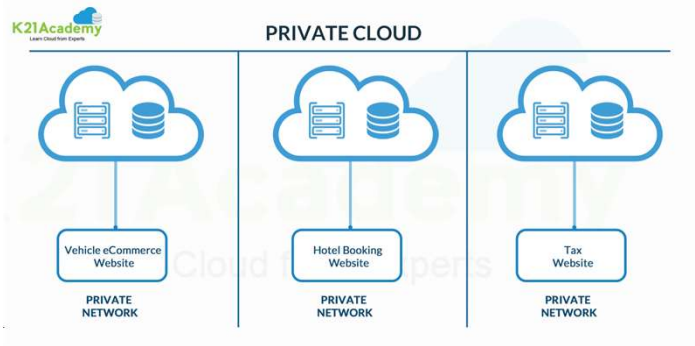
- Public Cloud Deployment model Providers:
- AWS
- Google App Engine
- Microsoft Azure
- IBM Cloud



Cloud Computing

NIST 5–3–4 Rule/Principle of Cloud Computing

- Private Cloud Deployment model Providers:
- Amazon Web Services
- Microsoft Azure
- Google Cloud Platform
- Dell
- Cisco

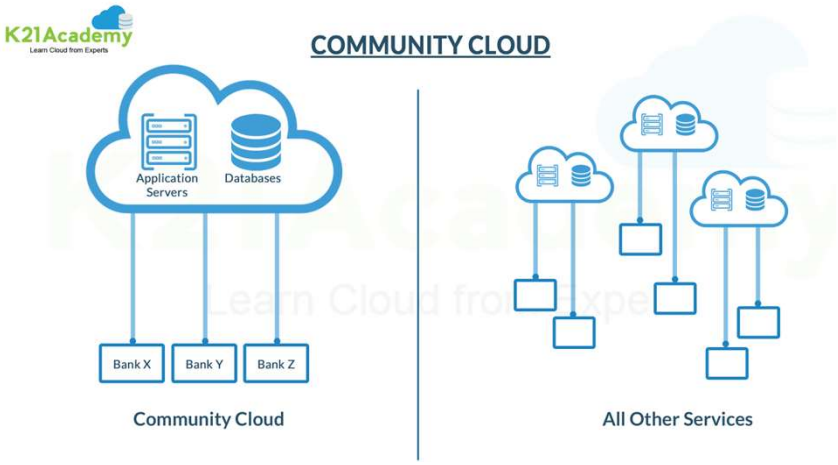


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Cloud Computing

NIST 5–3–4 Rule/Principle of Cloud Computing

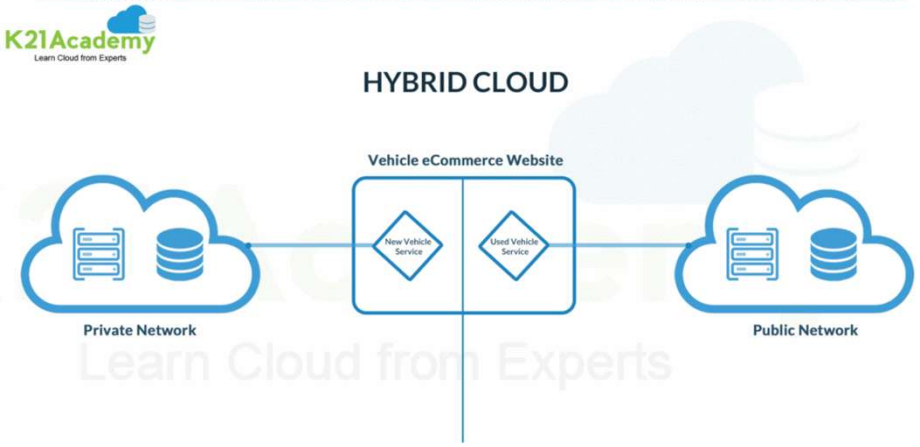


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Cloud Computing

NIST 5–3–4 Rule/Principle of Cloud Computing



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Cloud Computing

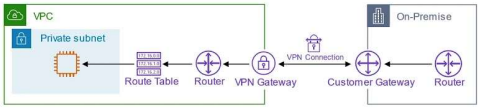
NIST 5–3–4 Rule/Principle of Cloud Computing

Cloud Deployment Models

Cloud-Native
Everything built on the Cloud



Hybrid-Architecture
Using both **On-Premise** and the **Cloud**



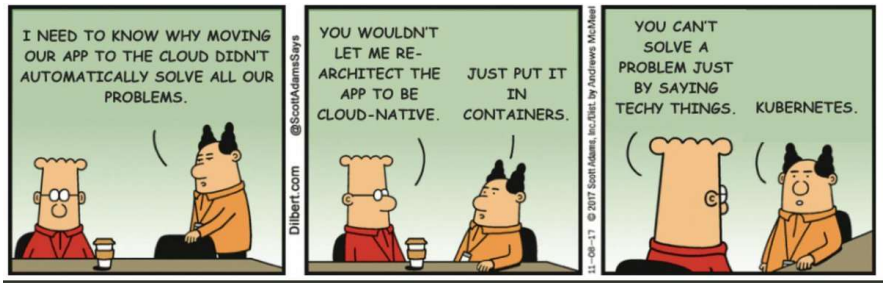
Cross-Cloud
Using **Multiple Cloud Providers**
Aka multi-cloud, hybrid-cloud



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Cloud Computing – All good news?



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Cloud Computing – All good news?

when our scale required building our own dedicated storage infrastructure, we named the project "Magic Pocket." Two and a half years later, we're excited to announce that we're now storing and serving over 90% of our users' data on our custom-built infrastructure.

Looking at the bigger picture, ***S3 was simply an expensive*** *default choice among many competitors—including our own Magic Pocket block store.*

https://dropbox.tech/infrastructure/magic-pocket-infrastructure

https://dropbox.tech/infrastructure/abstracting-cloud-storage-backends-with-object-store

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Cloud Computing – All good news?

The cloud excels at two *ends of the spectrum, where only one end was ever relevant for us. The* ***first end is when your application is so simple*** *and low traffic that you really do save on complexity by starting with fully managed services. (…) It remains a fabulous way to get started when you have no customers, and it'll carry you quite far even once you start having some.*

(…)

The second is when your load is highly irregular*. When you have wild swings or towering peaks in usage. When the baseline is a sliver of your largest needs. Or when you have no idea whether you need ten servers or a hundred. There's nothing like the cloud when that happens, like we learned when launching HEY, and suddenly 300,000 users signed up to try our service in three weeks instead of our forecast of 30,000 in six months.*

https://world.hey.com/dhh/why-we-re-leaving-the-cloud-654b47e0

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Cloud Computing – All good news?

Just over a year ago, we announced our intention to leave the cloud. We then shared our complete $3.2 million cloud budget for 2022, (…)

A month later, we placed an order for $600,000 worth of Dell servers to carry our exit, and did the math to conservatively estimate $7 million in savings over the next five years.

https://world.hey.com/dhh/the-big-cloud-exit-faq-20274010

Enter serverless. It's the same financial mechanics as above, but you can slice the server far more thinly. Instead of renting out your one big server to seven customers at $200/month, you rent out individual function executions to 100 customers at $20/month. This now clears $1,000/month in profit instead of just $400/month. No wonder cloud providers love serverless!

https://world.hey.com/dhh/don-t-be-fooled-by-serverless-776cd730

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- https://www.datacenterdynamics.com/en/news/research-larger-data-centers-make-considerable-savings-on-operating-costs/

3. Cloud Keywords Overview

"Everything fails, all the time."

Werner Vogels, CTO, Amazon.com

Ricardo Filipe, ISEC (ricardo.filipe@isec.pt)

Key Concepts

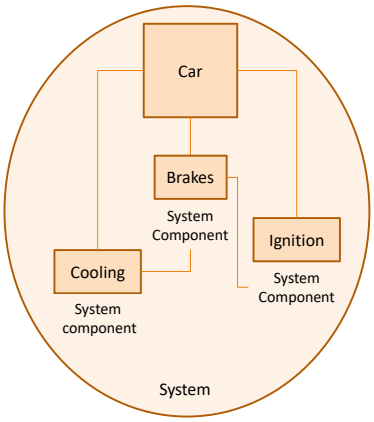


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Reliability

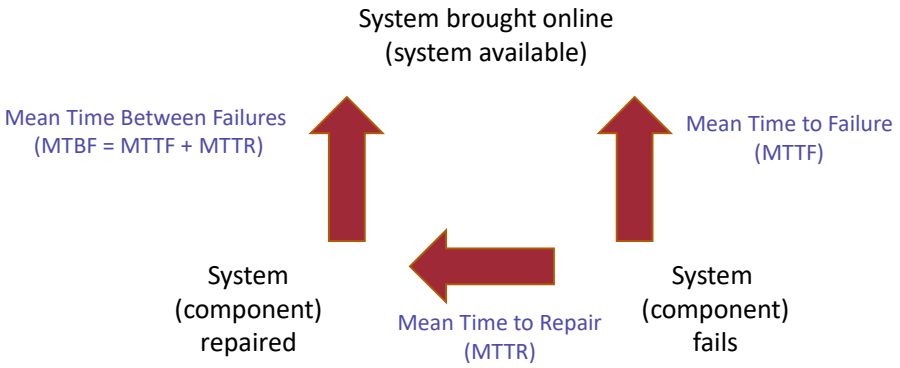
- A measure of your system's ability to provide functionality when desired by the user.
- System includes all system components: hardware, firmware, and software.
- Probability that your entire system will function as intended for a specified period.
- Mean time between failures (MTBF) = total time in service/number of failures



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Understanding reliability metrics



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Availability

- Normal operation time / total time
- A percentage of uptime (for example, 99.9 percent) over time (for example, 1 year)
- Number of 9s – Five 9s means 99.999 percent availability

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High availability

- System can withstand some measure of degradation while still remaining available.
- Downtime is minimized.
- Minimal human intervention is required.



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Availability tiers

Availability	Max Disruption (per year)	Application Category
99%	3 days 15 hours	Batch processing, data extraction, transfer, and load jobs
99.9%	8 hours 45 minutes	Internal tools like knowledge management, project tracking
99.95%	4 hours 22 minutes	Online commerce, point of sale
99.99%	52 minutes	Video delivery, broadcast systems
99.999%	5 minutes	ATM transactions, telecommunications systems

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Factors that influence availability

Recoverability

- The process, policies, and procedures that are related to restoring service after a catastrophic event.

Fault tolerance

- The built-in redundancy of an application's components and its ability to remain operational.

Scalability

- The ability of an application to accommodate increases in capacity needs without changing design.

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What is Total Cost of Ownership (TCO)?

Total Cost of Ownership (TCO) is the financial estimate to help identify direct and indirect costs of a system.

Why use TCO?

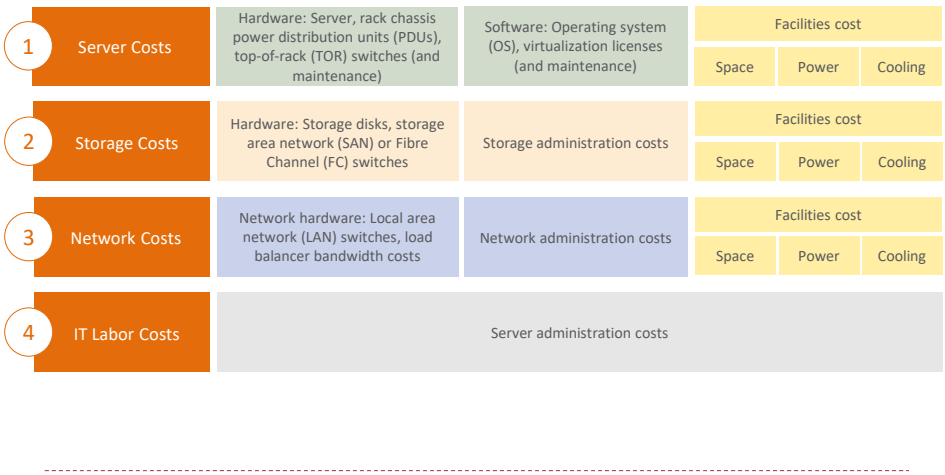
- To compare the costs of running an **entire infrastructure environment or specific workload** on-premises versus on AWS
- To budget and **build the business case** for moving to the cloud



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TCO considerations



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Other concepts…

- Cost Savings (TCO)
  - Infrastructure cost savings/ avoidance from moving to the cloud.
- Staff Productivity
  - Efficiency improvement by function on a task-by-task basis.
  - Example: Over 500 hours per year of server configuration time saved
- Business Agility
  - Deploying new features/ applications faster and reducing errors
- Operational Resilience
  - Benefit of improved availability, security, and compliance.
  - Example
    - Critical workloads run in multiple AZs and Regions for robust DR.
- Sustainability
  - Environmental sustainability is a shared responsibility between customers and AWS
  - https://aws.amazon.com/blogs/architecture/how-to-select-a-region-for-your-workload-based-on-sustainability-goals/

64

Bibliography (always in progress...)

- Based on AWS Academy
- Several papers.

65

4. AWS Global Infrastructure

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AWS Global Infrastructure

- The **AWS Global Infrastructure** is designed and built to deliver a **flexible**, **reliable**, **scalable**, and **secure** cloud computing environment with high-quality **global network performance**.
- AWS continually updates its global infrastructure footprint. Visit one of the following web pages for current infrastructure information:
  - AWS Global Infrastructure Map: https://aws.amazon.com/about-aws/global-infrastructure/#AWS_Global_Infrastructure_Map
    Choose a circle on the map to view summary information about the Region represented by the circle.
  - Regions and Availability Zones: https://aws.amazon.com/about-aws/global-infrastructure/regions_az/
    Choose a tab to view a map of the selected geography and a list of Regions, Edge locations, Local zones, and Regional Caches.



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AWS Regions

- An **AWS Region** is a geographical area.
  - Data replication across Regions is controlled by you.
  - Communication between Regions uses AWS backbone network infrastructure.
- Each Region provides full redundancy and connectivity to the network.
- A Region typically consists of two or more **Availability Zones**.

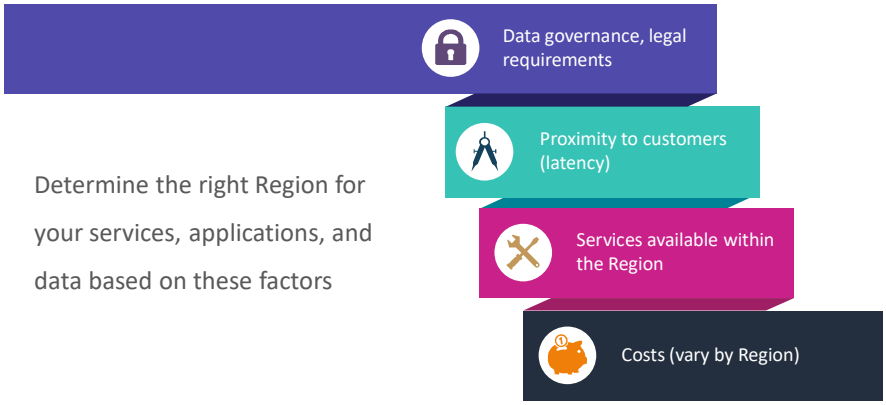


Example: London Region

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Selecting a Region

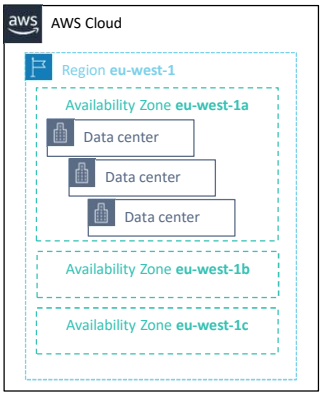
Determine the right Region for your services, applications, and data based on these factors



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Availability Zones

- Each **Region** has multiple Availability Zones.
- Each **Availability Zone** is a fully isolated partition of the AWS infrastructure.
  - Availability Zones consist of discrete **data centers**
  - They are designed for fault isolation
  - They are interconnected with other Availability Zones by using high-speed private networking
  - You choose your Availability Zones.
  - **AWS recommends replicating data and resources across Availability Zones** for resiliency.



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AWS data centers

- AWS data centers are designed for security.
- Data centers are where the data resides and data processing occurs.
- Each data center has redundant power, networking, and connectivity, and is housed in a separate facility.
- A data center typically has 50,000 to 80,000 physical servers.

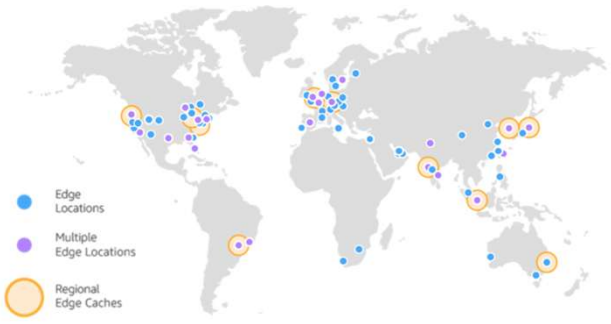


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Points of Presence

- AWS provides a global network of Points of Presence locations
- Consists of **edge locations** and a much smaller number of **Regional edge caches**
- Used with Amazon CloudFront
  - A global Content Delivery Network (CDN), that delivers content to end users with reduced latency
- Regional edge caches used for content with infrequent access.

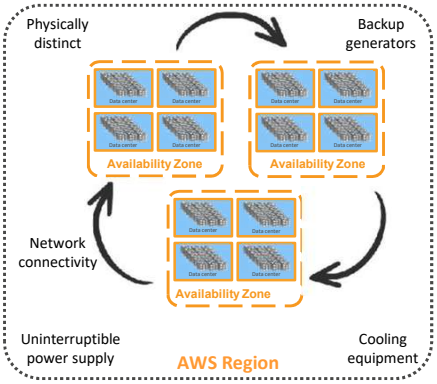


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AWS infrastructure features

- Elasticity and scalability
  - Elastic infrastructure; dynamic adaption of capacity
  - Scalable infrastructure; adapts to accommodate growth
- Fault-tolerance
  - Continues operating properly in the presence of a failure
  - Built-in redundancy of components
- High availability
  - High level of operational performance
  - Minimized downtime
  - No human intervention



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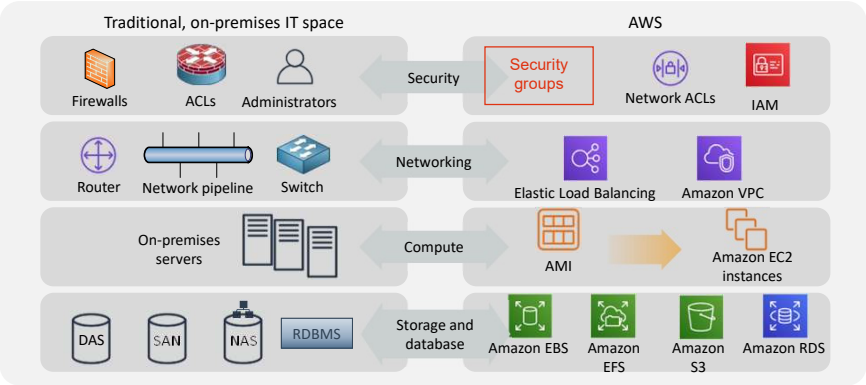
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5. Amazon Web Services (AWS) – General Information

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Similarities between AWS and traditional IT



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What is AWS?

- AWS is a **secure cloud platform** that offers a **broad set of global cloud-based products**.
- AWS provides you with **on-demand access** to compute, storage, network, database, and other IT resources and management tools.
- AWS offers **flexibility**.
- You **pay only for the individual services you need**, for **as long as you use them**.
- AWS services **work together** like building blocks.

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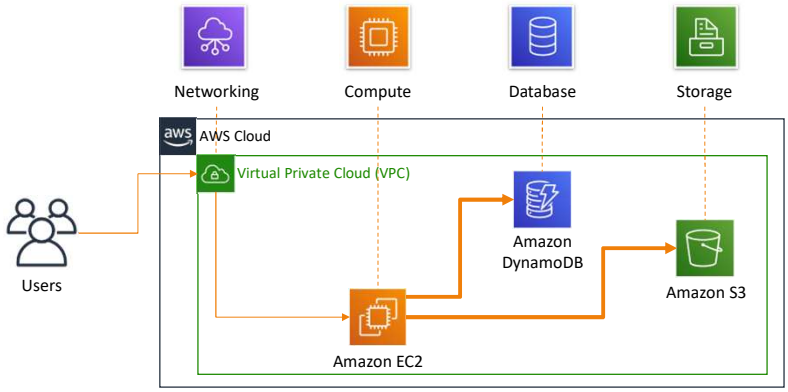
Categories of AWS services



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Simple solution example

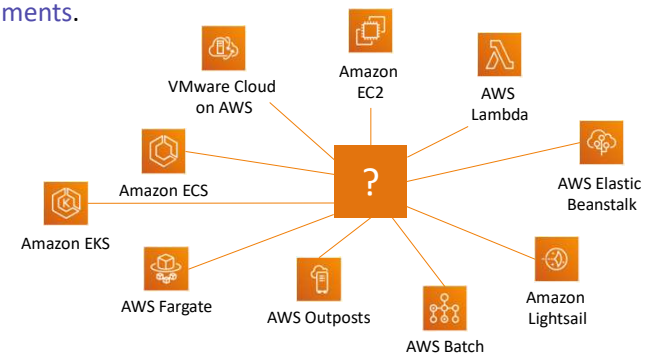


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Choosing a service

The service you select depends on your business goals and technology requirements.



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Services covered in this course

Compute services –
- Amazon EC2
- AWS Lambda
- AWS Elastic Beanstalk
- Amazon EC2 Auto Scaling

Storage services –
- Amazon S3

Management and Governance services –
- AWS CloudWatch
- AWS Auto Scaling
- AWS Command Line Interface
- AWS Management Console

Database services –
- Amazon RDS
- Amazon DynamoDB
- Amazon Redshift
- Amazon Aurora

Security, Identity, and Compliance services –
- AWS IAM

Internet of Things
- AWS IoT Core

Networking and Content Delivery services –
- Amazon VPC
- Elastic Load Balancing
- Amazon SNS

AWS Cost Management services –
- AWS Cost & Usage Report
- AWS Budgets
- AWS Cost Explorer

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Three ways to interact with AWS



AWS Management Console

Easy-to-use graphical interface



Command Line Interface (AWS CLI)

Access to services by discrete commands or scripts



Software Development Kits (SDKs)

Access services directly from your code (such as Java, Python, and others)

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AWS pricing model

Three fundamental drivers of cost with AWS

Compute

- Charged per hour/second*
- Varies by instance type

*Linux only

Storage

- Charged typically per GB

Data transfer

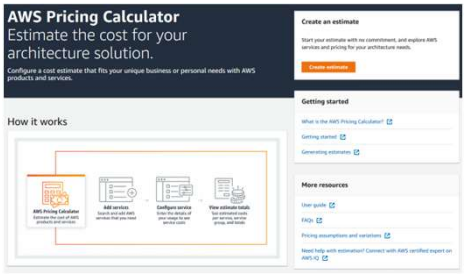
- Outbound is aggregated and charged
- Inbound has no charge (with some exceptions)
- Charged typically per GB

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AWS Pricing Calculator

Use the AWS Pricing Calculator to:

- Estimate monthly costs
- Identify opportunities to reduce monthly costs
- Model your solutions before building them
- Explore price points and calculations behind your estimate
- Find the available instance types and contract terms that meet your needs
- Name your estimate and create and name groups of services



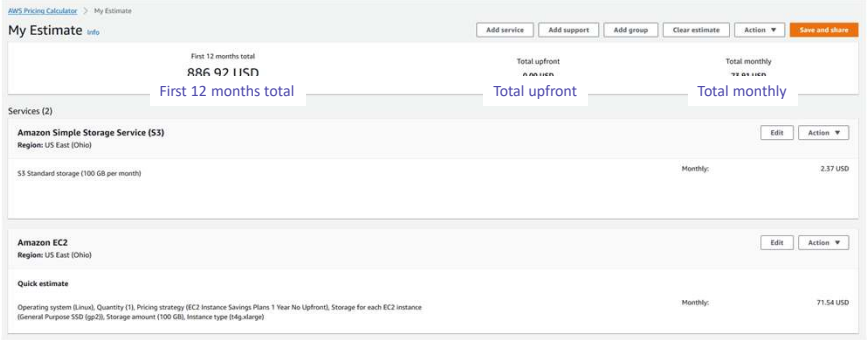
Access the AWS Pricing Calculator

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Reading an estimate

Your estimate is broken into: first 12 months total, total upfront, and total monthly.

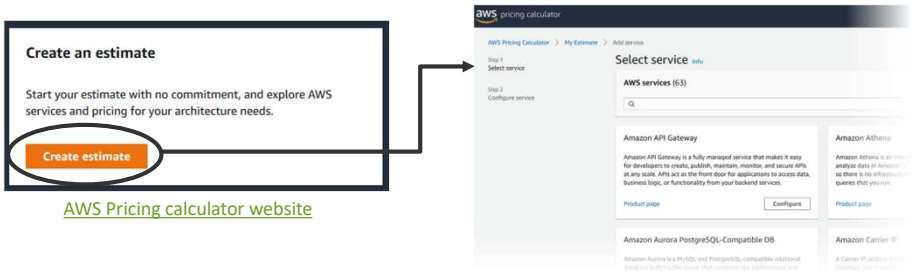


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Activity: AWS Pricing Calculator activity

- Break up into groups of four or five and use the AWS Pricing Calculator and specifications provided to develop a cost estimate.
- Be prepared to report your findings back to the class.



AWS Pricing calculator website

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Activity: AWS Pricing Calculator activity

- Estimate service with:
  - an **Amazon API Gateway**
    - 1M request per month
    - Size of each request of 3M
  - A PostGresSQL
    - 1 node
    - Single AZ
    - Compare distinct Instance Type

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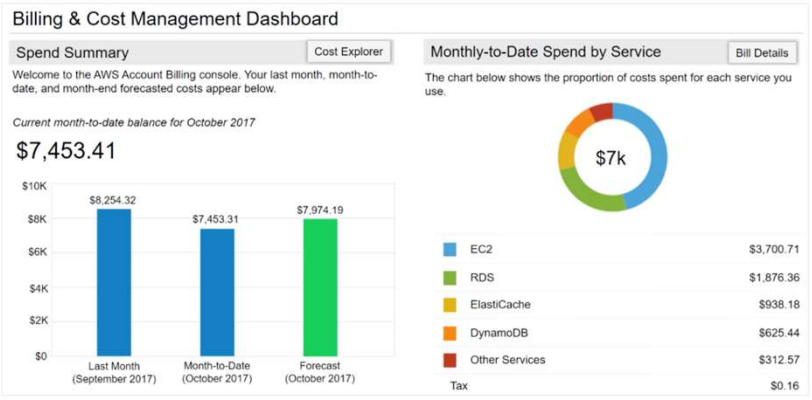
Introducing AWS Billing and Cost Management



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AWS Billing Dashboard



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Monthly bills

BILLS | COST EXPLORER | BUDGETS | REPORTS

Total		$7,453.41 USD
AWS Marketplace Charges		**$15.00**
▼ Usage Charges and Recurring Fees		$15.00
Invoice 32342548 – AWS Service Charges: Usage charge for this statement period	2017-10-10	$15.00
AWS Service Charges		**$7,438.41**
▼ Usage Charges and Recurring Fees		$7,414.41
Invoice 32342513 – AWS Service Charges: Usage charge for this statement period	2017-10-10	$7,414.41
▼ Usage Charges and Recurring Fees		$24.00
Invoice 32342507 – AWS Service Charges: Subscription charge	2017-10-10	$24.00

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Cost and usage reporting

BILLS | COST EXPLORER | BUDGETS | REPORTS

Product Code	Usage Type	Operation	Availability Zone	Usage Amount	Currency Code	Line Item Description
Amazon S3	Requests – Tier 1	ListAllMyBuckets		2	USD	$0.00 per request – PUT, COPY, POST, LIST under the global free tier
Amazon EC2	USW2-Boxusage:t2.micro	Runinstnaces:0002	us-west-2a	1	USD	$0.00 per Windows t2.micro instance-hour under monthly free tier
Amazon S3	Requests – Tier 1	ListAllMyBuckets		2	USD	$0.00 per request – PUT, COPY, POST, LIST under the global free tier
Amazon EC2	USW2-Boxusage:t2.micro	Runinstnaces:0002	us-west-2a	1	USD	$0.00 per Windows t2.micro instance-hour under monthly free tier
Amazon S3	Requests – Tier 1	ListAllMyBuckets		2	USD	$0.00 per request – PUT, COPY, POST, LIST under the global free tier
Amazon S3	Requests – Tier 1	ListAllMyBuckets		2	USD	$0.00 per request – PUT, COPY, POST, LIST under the global free tier

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Recorded demo: Amazon Billing dashboard



Billing dashboard demo at https://aws-tc-largeobjects.s3-us-west-2.amazonaws.com/ILT-TF-100-ACFNDS-20-EN/Module_2_Billing+Dashboard+v1.0.mp4.

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Support plans

AWS Support offers four support plans:

- **Basic Support** – Resource Center access, Service Health Dashboard, product FAQs, discussion forums, and support for health checks
- **Developer Support**: Support for early development on AWS
- **Business Support**: Customers that run production workloads
- **Enterprise Support**: Customers that run business and mission-critical workloads



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LR3

Case severity and response times

	Critical	Urgent	High	Normal	Low
Basic			No Case Support		
Developer Plan (Business hours)				12 hours or less	24 hours or less
Business Plan (24/7)		1 hour or less	4 hours or less	12 hours or less	24 hours or less
Enterprise Plan (24/7)	15 minutes or less	1 hour or less	4 hours or less	12 hours or less	24 hours or less

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Monitoring AWS resources

To use AWS efficiently, you need insight into your AWS resources:

- How do you know when you should **launch more Amazon EC2 instances**?
- Is your **application's performance or availability** being affected by a lack of sufficient capacity?
- How much of your infrastructure is actually **being used**?

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Slide 93

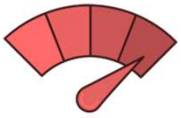
LR3 Check paragraph spacing on the Notes page -- different from other pages.
Libby Riker; 16/05/2022

Amazon CloudWatch



Amazon CloudWatch





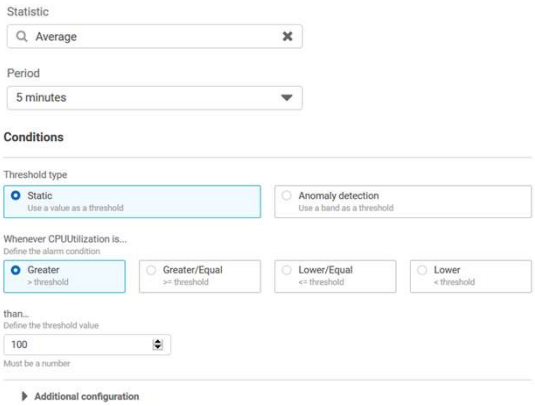
- Monitors –
  - AWS resources
  - Applications that run on AWS
- Collects and tracks –
  - Standard metrics
  - Custom metrics
- Alarms –
  - Send notifications to an Amazon SNS topic
  - Perform Amazon EC2 Auto Scaling or Amazon EC2 actions
- Events –
  - Define rules to match changes in AWS environment and route these events to one or more target functions or streams for processing

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CloudWatch alarms

- Create alarms based on –
  - Static threshold
  - Anomaly detection
  - Metric math expression
- Specify –
  - Namespace
  - Metric
  - Statistic
  - Period
  - Conditions
  - Additional configuration
  - Actions



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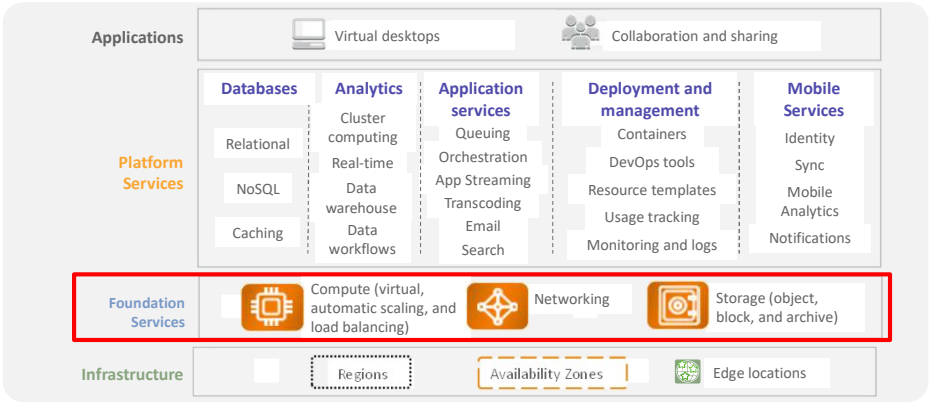
96

6. AWS Services

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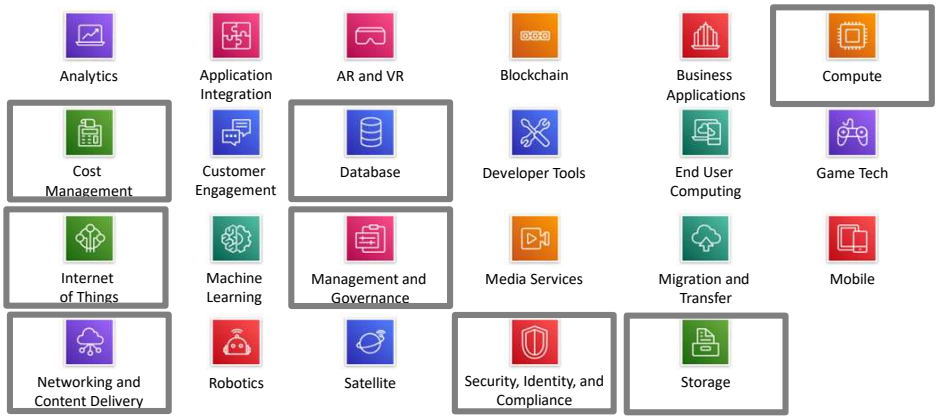
AWS foundational services



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AWS categories of services



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Storage service category



Photo from https://www.pexels.com/photo/black-and-grey-device-159282/

AWS storage services

- Amazon Simple Storage Service (Amazon S3)
- Amazon Elastic Block Store (Amazon EBS)
- Amazon Elastic File System (Amazon EFS)
- Amazon Simple Storage Service Glacier

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Compute service category

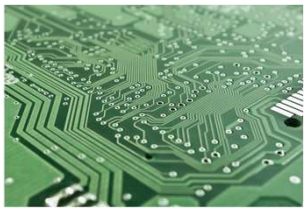


Photo from https://www.pexels.com/photo/technology-computer-lines-board-50711/

AWS Compute services

- Amazon EC2
- Amazon EC2 Auto Scaling
- Amazon Elastic Container Service (Amazon ECS)
- Amazon EC2 Container Registry
- AWS Elastic Beanstalk
- AWS Lambda
- Amazon Elastic Kubernetes Service (Amazon EKS)
- AWS Fargate

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Database service category



Photo from https://aws.amazon.com/compliance/data-center/data-centers/

AWS Database services

- Amazon Relational Database Service
- Amazon Aurora
- Amazon Redshift
- Amazon DynamoDB

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Networking and content delivery service category



Photo by Umberto on Unsplash

AWS networking and content delivery services

- Amazon VPC
- Elastic Load Balancing
- Amazon CloudFront
- AWS Transit Gateway
- Amazon Route 53
- AWS Direct Connect
- AWS VPN

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Security, identity, and compliance service category



Photo by Paweł Czerwiński on Unsplash

AWS security, identity, and compliance services

- AWS Identity and Access Management (IAM)
- AWS Organizations
- Amazon Cognito
- AWS Artifact
- AWS Key Management Service
- AWS Shield

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AWS cost management service category

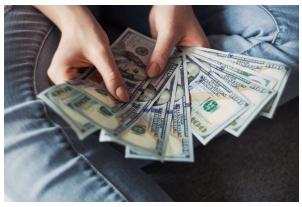


Photo by Alexander Mils on Unsplash

AWS cost management services

- AWS Cost and Usage Report
- AWS Budgets
- AWS Cost Explorer

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Management and governance service category

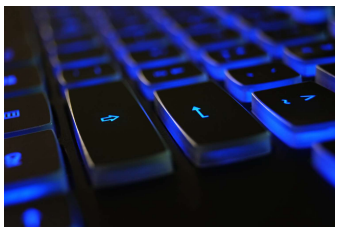


Photo by Marta Branco from Pexels

AWS management and governance services

- AWS Management Console
- AWS Config
- Amazon CloudWatch
- AWS Auto Scaling
- AWS Command Line Interface
- AWS Trusted Advisor
- AWS Well-Architected Tool
- AWS CloudTrail

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Activity: AWS Management Console clickthrough



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Hands-on activity: AWS Management Console clickthrough

1. Launch the Sandbox hands-on environment and connect to the AWS Management Console**.**
2. Explore the AWS Management Console.
   - A. Click the **Services** menu.
   - B. Notice how services are grouped into service categories. For example, the **EC2** service appears in the **Compute** service category.
     - Question #1: Under which service category does the **IAM** service appear?
     - Question #2: Under which service category does the **Amazon VPC** service appear?
   - C. Click the **Amazon VPC** service. Notice that the dropdown menu in the top-right corner displays an AWS Region (for example, it might display *N. Virginia*).
   - D. Click the Region menu and switch to a different Region. For example, choose **EU (London)**.
   - E. Click **Subnets** (on the left side of the screen). The Region has three subnets in it. Click the box next to one of the subnets. Notice that the bottom half of the screen now displays details about this subnet.
     - Question #3: Does the subnet you selected exist at the level of the Region or at the level of the Availability Zone?
   - F. Click **Your VPCs**. An existing VPC is already selected.
     - Question #4: Does the VPC exist at the level of the Region or the level of the Availability Zone?
     - Question #5: Which services are global instead of Regional? Check Amazon EC2, IAM, Lambda, and Route 53.

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Activity answer key

- Question #1: Under which service category does the **IAM** service appear?
  - **Answer**: **Security, Identity, & Compliance**.
- Question #2: Under which service category does the **Amazon VPC** service appear?
  - **Answer**: **Networking & Content Delivery**
- Question #3: Does the subnet that you selected exist at the level of the Region or the level of the Availability Zone?
  - **Answer**: Subnets exist at the **level of the Availability Zone.**
- Question #4: Does the VPC exist at the level of the Region or the level of the Availability Zone?
  - **Answer**: VPCs exist at the **Region level.**
- Question #5: Which of the following services are global instead of Regional? Check Amazon EC2, IAM, Lambda, and Route 53.
  - **Answer: IAM and Route 53 are global**. Amazon EC2 and Lambda are Regional.

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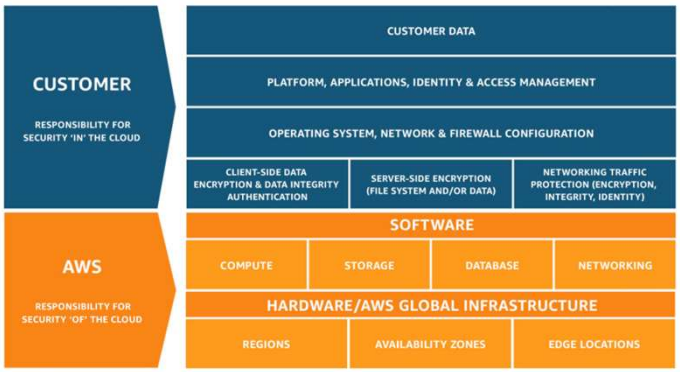
109

7. AWS Security

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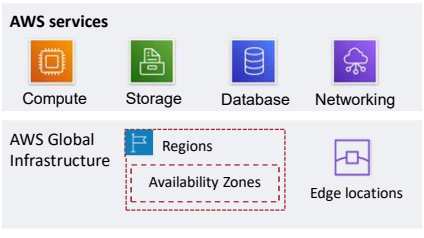
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AWS shared responsibility model



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AWS responsibility: Security *of* the cloud



AWS responsibilities:



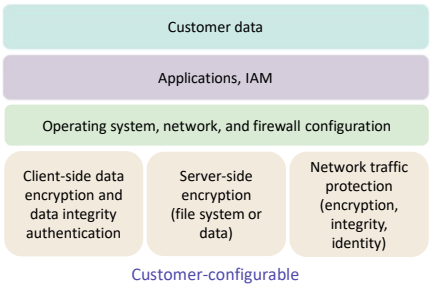
- Physical security of data centers
  - Controlled, need-based access
- Hardware and software infrastructure
  - Storage decommissioning, host operating system (OS) access logging, and auditing
- Network infrastructure
  - Intrusion detection



- Virtualization infrastructure
  - Instance isolation

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Customer responsibility: Security *in* the cloud



Customer responsibilities:

- Amazon Elastic Compute Cloud (Amazon EC2) instance operating system
  - Including patching, maintenance
- Applications
  - Passwords, role-based access, etc.
- Security group configuration
- OS or host-based firewalls
  - Including intrusion detection or prevention systems
- Network configurations
- Account management
  - Login and permission settings for each user

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AWS Identity and Access Management (IAM)

- Use **IAM** to manage access to **AWS resources** –
  - A resource is an entity in an AWS account that you can work with
  - Example resources; An Amazon EC2 instance or an Amazon S3 bucket
- *Example* – Control who can terminate Amazon EC2 instances
- Define fine-grained access rights –
  - **Who** can access the resource
  - **Which** resources can be accessed and what can the user do to the resource
  - **How** resources can be accessed
- IAM is a no-cost AWS account feature



AWS Identity and Access Management (IAM)

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IAM: Essential components


IAM user

A **person** *or* **application** that can authenticate with an AWS account.


IAM group

A **collection of IAM users** that are granted identical authorization.


IAM policy

The document that defines **which resources can be accessed** and the **level of access** to each resource.


IAM role

Useful mechanism to grant a set of permissions for making AWS service requests.

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Authenticate as an IAM user to gain access

When you define an **IAM user**, you select what ***types of access*** the user is permitted to use.

Programmatic access

- Authenticate using:
  - *Access key ID*
  - *Secret access key*
- Provides AWS CLI and AWS SDK access


AWS CLI


AWS Tools and SDKs

AWS Management Console access

- Authenticate using:
  - *12-digit Account ID or alias*
  - *IAM user name*
  - *IAM password*
- If enabled, **multi-factor authentication (MFA)** prompts for an authentication code.

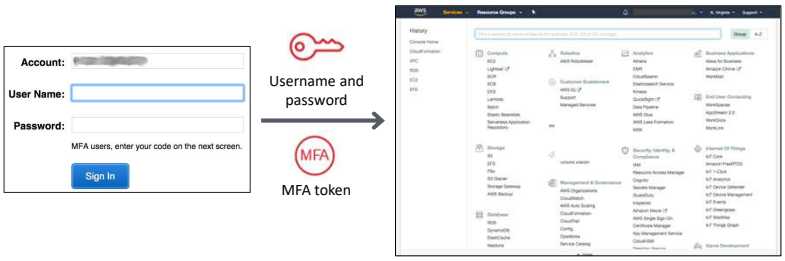

AWS Management Console

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IAM MFA

- MFA provides increased security.
- In addition to **username** and **password**, MFA requires a unique authentication code to access AWS services.



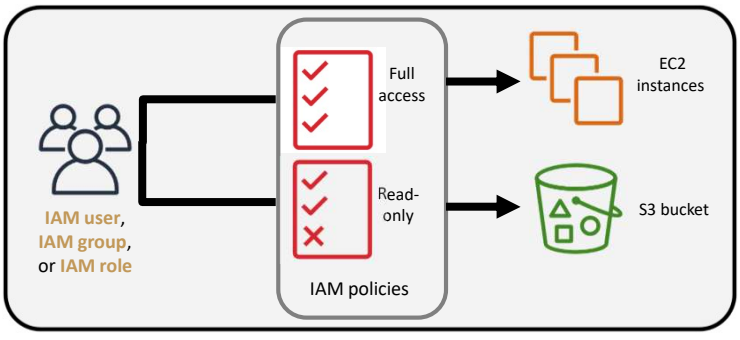
AWS Management Console

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Authorization: What actions are permitted

After the user or application is connected to the AWS account, what are they allowed to do?



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IAM: Authorization

- Assign permissions by creating an IAM policy.
- Permissions determine **which resources and operations** are allowed:
  - All permissions are implicitly denied by default.
  - If something is explicitly denied, it is never allowed.

Best practice: Follow the **principle of least privilege**.



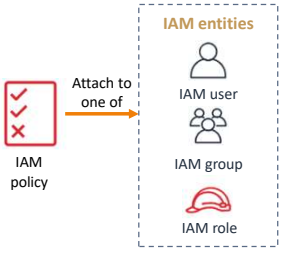
IAM permissions

Note: The scope of IAM service configurations is global. Settings apply across all AWS Regions.

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IAM policies

- **An IAM policy is a document that defines permissions**
  - Enables fine-grained access control
- Two types of policies – *identity-based* and *resource-based*
- **Identity-based** policies –
  - Attach a policy to any IAM entity
    - An IAM user, an IAM group, or an IAM role
  - Policies specify:
    - Actions that ***may*** be performed by the entity
    - Actions that ***may not*** be performed by the entity
  - A single *policy* can be attached to multiple *entities*
  - A single *entity* can have multiple *policies* attached to it
- **Resource-based** policies
  - Attached to a resource (such as an S3 bucket)



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IAM policy example

```
{
  "Version": "2012-10-17",
  "Statement":[{
    "Effect":"Allow",
    "Action":["DynamoDB:*","s3:*"],
    "Resource":[
        "arn:aws:dynamodb:region:account-number-without-hyphens:table/table-name",
      "arn:aws:s3:::bucket-name",
      "arn:aws:s3:::bucket-name/*"]
    },
    {
    "Effect":"Deny",
    "Action":["dynamodb:*","s3:*"],
    "NotResource":["arn:aws:dynamodb:region:account-number-without-hyphens:table/table-name",
      "arn:aws:s3:::bucket-name",
      "arn:aws:s3:::bucket-name/*"]
    }
  ]
}
```

Explicit allow gives users access to a specific DynamoDB table and…

…Amazon S3 buckets.

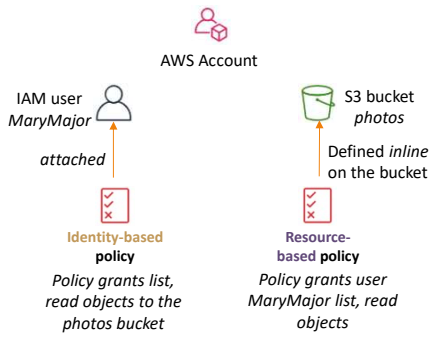
Explicit deny ensures that the users cannot use any other AWS actions or resources other than that table and those buckets.

An explicit deny statement **takes precedence** over an allow statement.

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Resource-based policies

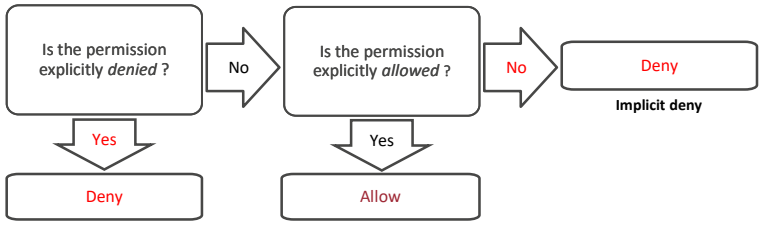
- *Identity-based policies* are attached to a user, group, or role
- **Resource-based policies** are attached to a resource (*not* to a user, group or role)
- Characteristics of resource-based policies –
  - Specifies who has access to the resource and what actions they can perform on it
  - The policies are *inline* only, not managed
- Resource-based policies are supported only by some AWS services



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IAM permissions

How IAM determines permissions:

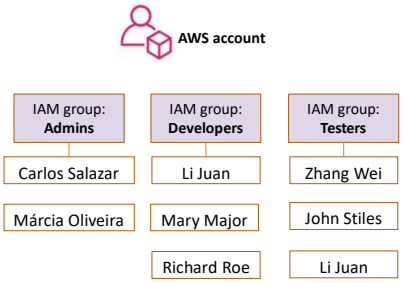


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IAM groups

- An **IAM group** is a collection of IAM users
- A group is used to grant the same permissions to multiple users
  - Permissions granted by attaching IAM *policy* or policies to the group
- A user can belong to multiple groups
- There is no default group
- Groups cannot be nested



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IAM roles

- An **IAM role** is an IAM identity with specific permissions
- Similar to an IAM user
  - Attach permissions policies to it
- Different from an IAM user
  - Not uniquely associated with one person
  - Intended to be *assumable* by a **person**, **application**, or **service**
- Role provides *temporary* security credentials
- Examples of how IAM roles are used to **delegate** access –
  - Used by an IAM user in the same AWS account as the role
  - Used by an AWS service—such as Amazon EC2—in the same account as the role
  - Used by an IAM user in a different AWS account than the role



IAM role

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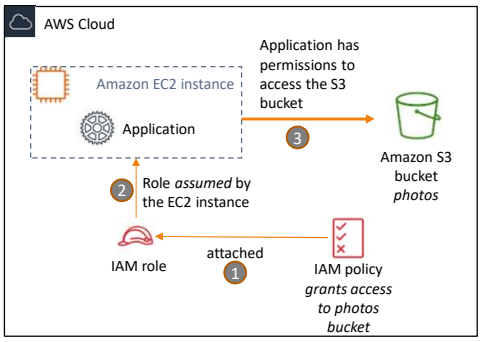
Example use of an IAM role

Scenario:

- An application that runs on an EC2 instance needs access to an S3 bucket

Solution:

- Define an IAM policy that grants access to the S3 bucket.
- Attach the policy to a role
- Allow the EC2 instance to assume the role



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Recorded demo: IAM

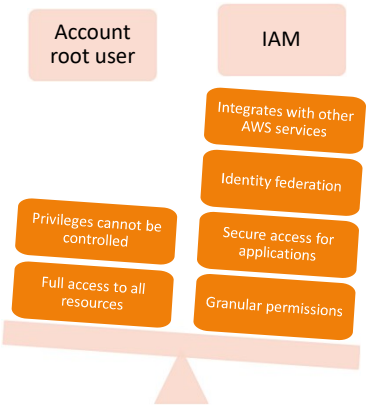


IAM Demo at https://aws-tc-largeobjects.s3-us-west-2.amazonaws.com/ILT-TF-100-ACFNDS-20-EN/Module_4_IAM+v2.0.mp4.

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AWS account root user access versus IAM access



- **Best practice**: Do not use the AWS account root user except when necessary.
  - Access to the **account root user** requires logging in with the *email address* (and password) that you used to create the account.
- Example actions that can only be done with the account root user:
  - Update the account root user password
  - Change the AWS Support plan
  - Restore an IAM user's permissions
  - Change account settings (for example, contact information, allowed Regions)

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8. AWS Networking and Content Delivery

Ricardo Filipe, ISEC (ricardo.filipe@isec.pt)

Amazon VPC



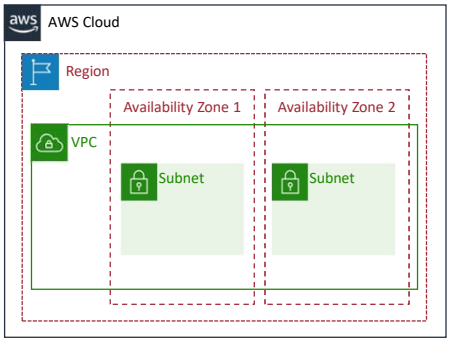
Amazon VPC

- Enables you to provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define
- Gives you control over your virtual networking resources, including:
  - Selection of IP address range
  - Creation of subnets
  - Configuration of route tables and network gateways
- Enables you to customize the network configuration for your VPC
- Enables you to use multiple layers of security

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VPCs and subnets

- VPCs:
  - Logically isolated from other VPCs
  - Dedicated to your AWS account
  - Belong to a single AWS Region and can span multiple Availability Zones
- Subnets:
  - Range of IP addresses that divide a VPC
  - Belong to a single Availability Zone
  - Classified as public or private



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Amazon Route 53

Amazon Route 53

- Is a highly available and scalable Domain Name System (DNS) web service
- Is used to route end users to internet applications by translating names (like *www.example.com*) into numeric IP addresses (like *192.0.2.1*) that computers use to connect to each other
- Is fully compliant with IPv4 and IPv6
- Connects user requests to infrastructure running in AWS and also outside of AWS
- Is used to check the health of your resources
- Features traffic flow
- Enables you to register domain names

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Amazon CloudFront



Amazon CloudFront

- Fast, global, and secure CDN service
- Global network of edge locations and Regional edge caches
- Self-service model
- Pay-as-you-go pricing

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9. AWS Databases

Ricardo Filipe, ISEC (ricardo.filipe@isec.pt)

Amazon Relational Database Service



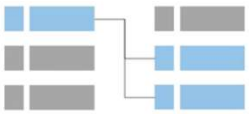
Amazon Relational Database Service (Amazon RDS)

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Challenges of relational databases

- Server maintenance and energy footprint
- Software installation and patches
- Database backups and high availability
- Limits on scalability
- Data security
- Operating system (OS) installation and patches

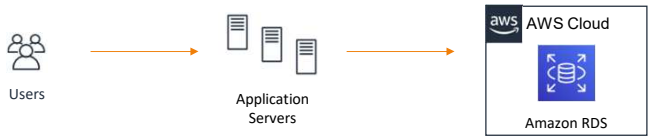


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Amazon RDS

Managed service that sets up and operates a relational database in the cloud.



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Amazon RDS DB instances

Amazon RDS

Amazon RDS DB main instance

DB Instance Class
- CPU
- Memory
- Network performance

DB Instance Storage
- Magnetic
- General Purpose (solid state drive, or SSD)
- Provisioned IOPS

MySQL

Amazon Aurora

Microsoft SQL Server

PostgreSQL

MariaDB

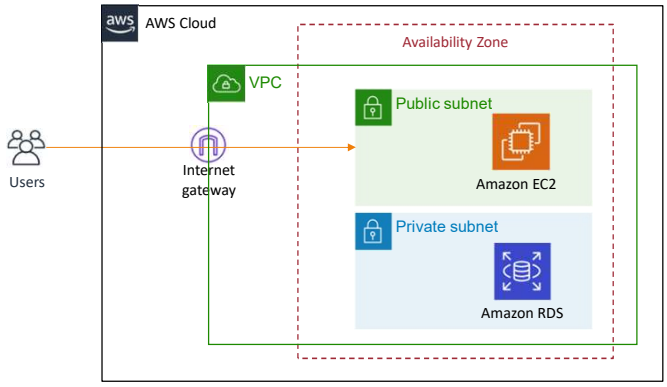
Oracle

DB engines

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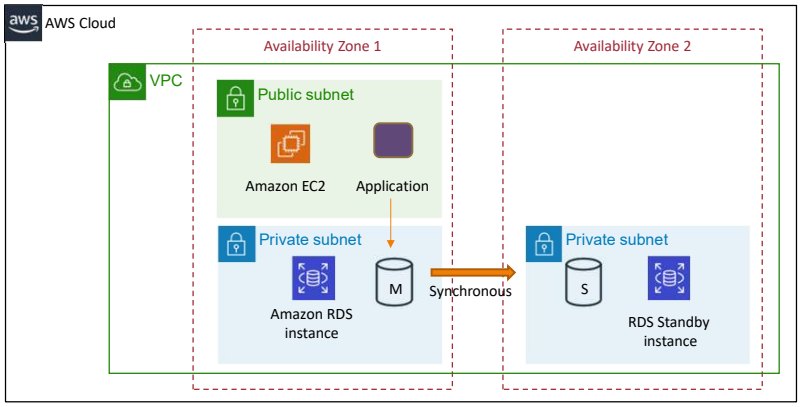
Amazon RDS in a virtual private cloud (VPC)



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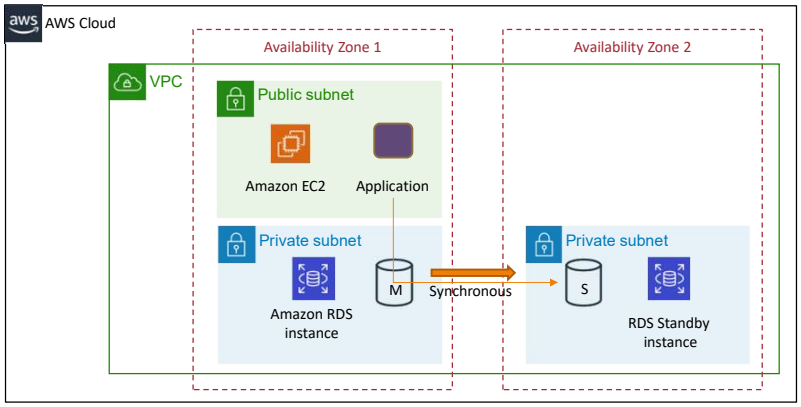
High availability with Multi-AZ deployment (1 of 2)



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High availability with Multi-AZ deployment (2 of 2)



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What is Amazon DynamoDB?

Fast and flexible NoSQL database service for any scale

Amazon DynamoDB

- NoSQL database tables
- Virtually unlimited storage
- Items can have differing attributes
- Low-latency queries
- Scalable read/write throughput

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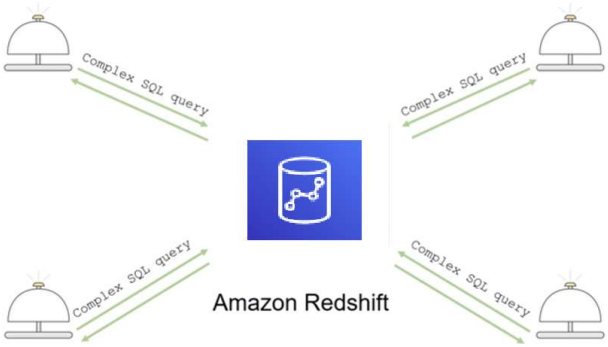
Amazon DynamoDB core components

- Tables, items, and attributes are the core DynamoDB components
- DynamoDB supports two different kinds of primary keys: Partition key and partition and sort key

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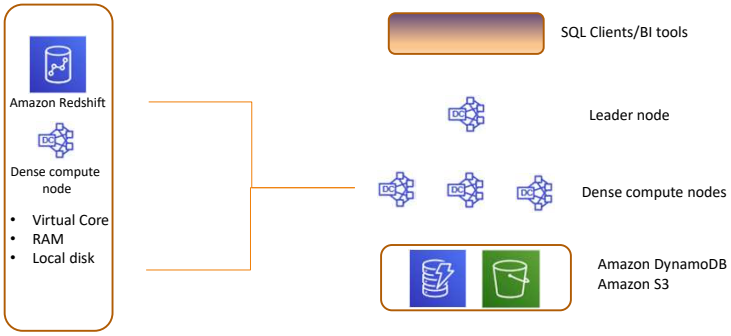
Introduction to Amazon Redshift



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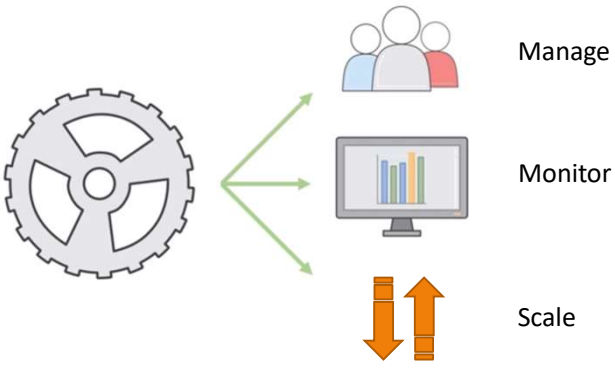
Parallel processing architecture



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Automation and scaling



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Amazon Redshift use cases (1 of 2)

- Enterprise data warehouse (EDW)
  - Migrate at a pace that customers are comfortable with
  - Experiment without large upfront cost or commitment
  - Respond faster to business needs
- Big data
  - Low price point for small customers
  - Managed service for ease of deployment and maintenance
  - Focus more on data and less on database management



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Amazon Aurora



Amazon Aurora

- Enterprise-class relational database
- Compatible with MySQL or PostgreSQL
- Automate time-consuming tasks (such as provisioning, patching, backup, recovery, failure detection, and repair).

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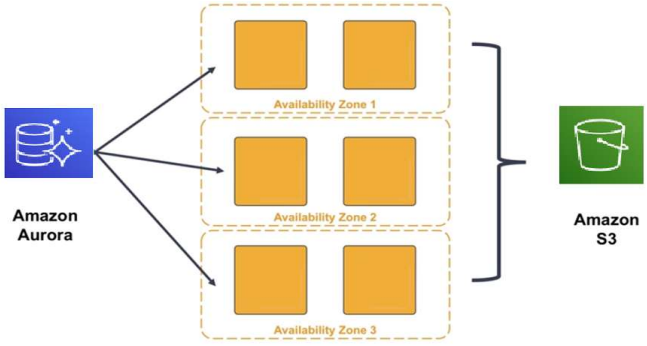
Amazon Aurora service benefits



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High availability



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10. AWS Compute

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Categorizing compute services

Services	Key Concepts	Characteristics	Ease of Use
• Amazon EC2	• Infrastructure as a service (IaaS) • Instance-based • **Virtual machines**	• Provision virtual machines that you can manage as you choose	A familiar concept to many IT professionals.
• AWS Lambda	• **Serverless** computing • Function-based • Low-cost	• Write and deploy code that runs on a schedule or that can be triggered by events • Use when possible (architect for the cloud)	A relatively new concept for many IT staff members, but easy to use after you learn how.
• Amazon ECS • Amazon EKS • AWS Fargate • Amazon ECR	• **Container-based** computing • Instance-based	• Spin up and run jobs more quickly	AWS Fargate reduces administrative overhead, but you can use options that give you more control.
• AWS Elastic Beanstalk	• Platform as a service (PaaS) • For **web applications**	• Focus on your code (building your application) • Can easily tie into other services—databases, Domain Name System (DNS), etc.	Fast and easy to get started.

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Amazon Elastic Compute Cloud (Amazon EC2)

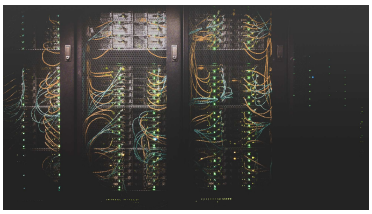
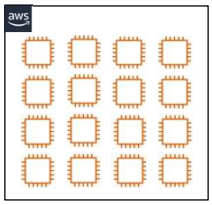


Photo by Taylor Vick on Unsplash

On-premises servers

Example uses of Amazon EC2 instances

- ✓ Application server
- ✓ Web server
- ✓ Database server
- ✓ Game server
- ✓ Mail server
- ✓ Media server
- ✓ Catalog server
- ✓ File server
- ✓ Computing server
- ✓ Proxy server



Amazon EC2 instances



Photo by panumas nikhomkhai from Pexels

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Amazon EC2 overview

Amazon EC2

- **Amazon Elastic Compute Cloud (Amazon EC2)**
  - Provides virtual machines—referred to as EC2 instances—in the cloud.
  - Gives you *full control* over the guest operating system (Windows or Linux) on each instance.
- You can launch instances of any size into an Availability Zone anywhere in the world.
  - Launch instances from **Amazon Machine Images (AMIs)**.
  - Launch instances with a few clicks or a line of code, and they are ready in minutes.
- You can control traffic to and from instances.

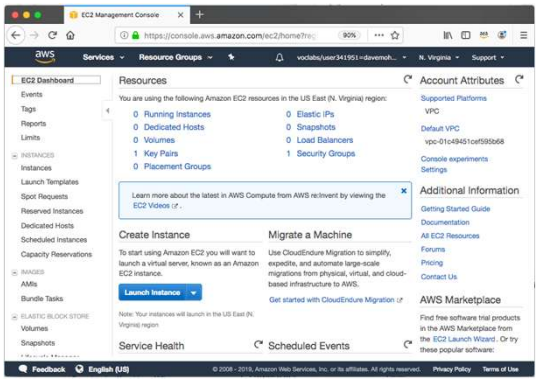
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Launching an Amazon EC2 instance

This section of the module walks through **nine key decisions** to make when you create an EC2 instance by using the AWS Management Console **Launch Instance Wizard.**

- Along the way, essential Amazon EC2 concepts will be explored.



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1. Select an AMI

Choices made using the Launch Instance Wizard:

1. **AMI**
2. **Instance Type**
3. **Network settings**
4. **IAM role**
5. **User data**
6. **Storage options**
7. **Tags**
8. **Security group**
9. **Key pair**



- Amazon Machine Image (AMI)
  - Is a template that is used to create an EC2 instance (which is a **virtual machine, or VM,** that runs in the AWS Cloud)
  - Contains a **Windows** or **Linux** operating system
  - Often also has some **software** pre-installed
- AMI choices:
  - Quick Start – *Linux and Windows AMIs that are provided by AWS*
  - My AMIs – *Any AMIs that you created*
  - AWS Marketplace – *Pre-configured templates from third parties*
  - Community AMIs – *AMIs shared by others; use at your own risk*

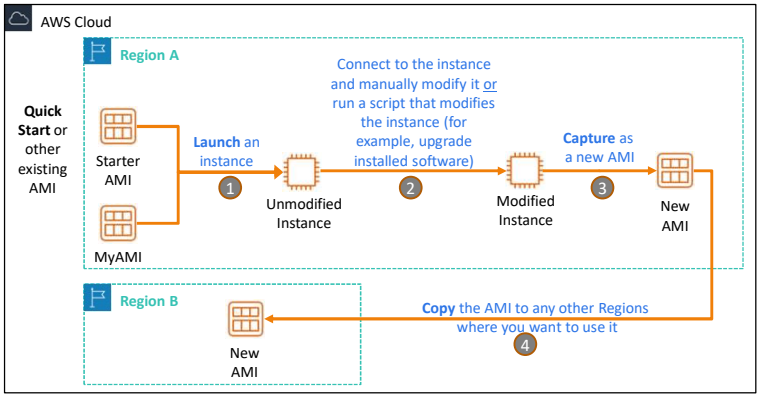
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Creating a new AMI: Example

AMI details

(Optional) Import a virtual machine



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2. Select an instance type

Choices made using the Launch Instance Wizard:

1. **AMI**
2. **Instance Type**
3. **Network settings**
4. **IAM role**
5. **User data**
6. **Storage options**
7. **Tags**
8. **Security group**
9. **Key pair**

- Consider your use case
  - How will the EC2 instance you create be used?
- The **instance type** that you choose determines –
  - Memory (RAM)
  - Processing power (CPU)
  - Disk space and disk type (Storage)
  - Network performance
- Instance type categories –
  - General purpose
  - Compute optimized
  - Memory optimized
  - Storage optimized
  - Accelerated computing
- Instance types offer *family*, *generation*, and *size*



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EC2 instance type naming and sizes

Instance type naming

- Example: **t3.large**
  - T is the family name
  - 3 is the generation number
  - Large is the size

Example instance sizes

Instance Name	vCPU	Memory (GB)	Storage
t3.nano	2	0.5	EBS-Only
t3.micro	2	1	EBS-Only
t3.small	2	2	EBS-Only
t3.medium	2	4	EBS-Only
t3.large	2	8	EBS-Only
t3.xlarge	4	16	EBS-Only
t3.2xlarge	8	32	EBS-Only

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Select instance type: Based on use case

	General Purpose	Compute Optimized	Memory Optimized	Accelerated Computing	Storage Optimized
Instance Types	a1, m4, m5, t2, t3	c4, c5	r4, r5, x1, z1	f1, g3, g4, p2, p3	d2, h1, i3
Use Case	Broad	High performance	In-memory databases	Machine learning	Distributed file systems

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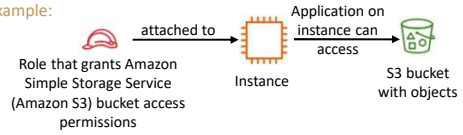
4. Attach IAM role (optional)

Choices made by using the Launch Instance Wizard:

1. **AMI**
2. **Instance Type**
3. **Network settings**
4. **IAM role**
5. **User data**
6. **Storage options**
7. **Tags**
8. **Security group**
9. **Key pair**

- Will software on the EC2 instance need to interact with other AWS services?
  - If yes, attach an appropriate **IAM Role**.
- An AWS Identity and Access Management (IAM) role that is attached to an EC2 instance is kept in an **instance profile**.
- You are *not* restricted to attaching a role only at instance launch.
  - You can also attach a role to an instance that already exists.

Example:

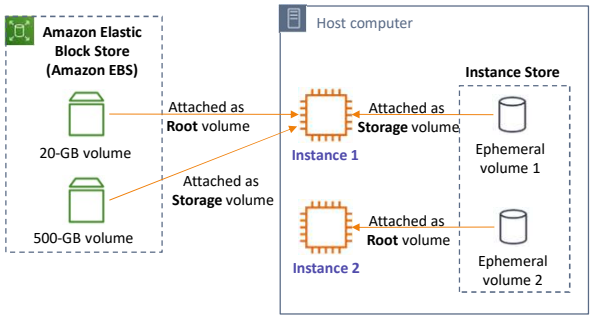


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Example storage options

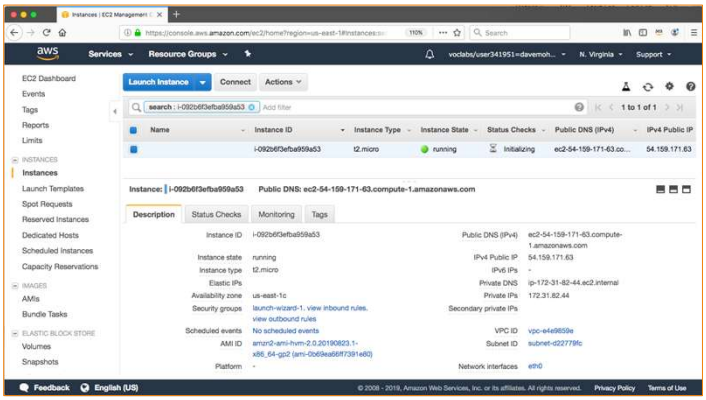
- **Instance 1** characteristics –
  - It has an **Amazon EBS** *root volume* type for the operating system.
  - What will happen if the instance is stopped and then started again?
- **Instance 2** characteristics –
  - It has an **Instance Store** *root volume* type for the operating system.
  - What will happen if the instance stops (because of user error or a system malfunction)?



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Amazon EC2 console view of a running EC2 instance



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Another option: Launch an EC2 instance with the AWS Command Line Interface

- EC2 instances can also be created programmatically.



AWS Command Line Interface (AWS CLI)

- This example shows how simple the command can be.
  - This command assumes that the key pair and security group already exist.
  - More options could be specified. See the AWS CLI Command Reference for details.

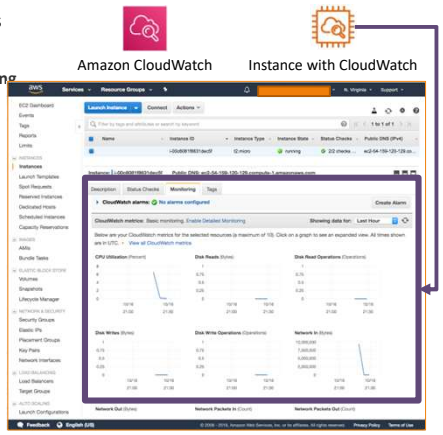
Example command:

```
aws ec2 run-instances \
--image-id ami-1a2b3c4d \
--count 1 \
--instance-type c3.large \
--key-name MyKeyPair \
--security-groups MySecurityGroup \
--region us-east-1
```

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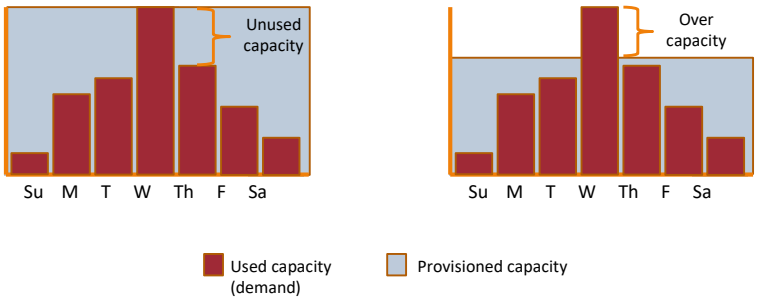
Amazon CloudWatch for monitoring

- Use **Amazon CloudWatch** to monitor EC2 instances
  - Provides near-real-time metrics
  - Provides charts in the Amazon EC2 console **Monitoring** tab that you can view
  - Maintains 15 months of historical data
- **Basic monitoring**
  - Default, no additional cost
  - Metric data sent to CloudWatch every 5 minutes
- **Detailed monitoring**
  - Fixed monthly rate for seven pre-selected metrics
  - Metric data delivered every 1 minute



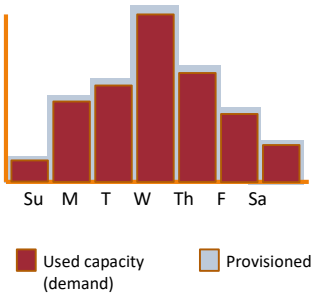
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Why is scaling important?



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Amazon EC2 Auto Scaling

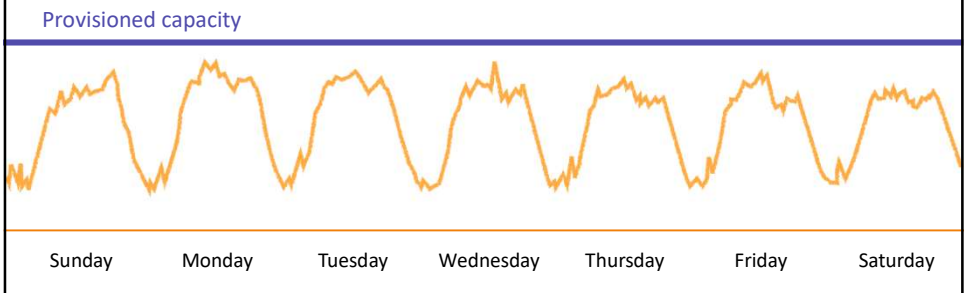


- Helps you maintain application availability
- Enables you to automatically add or remove EC2 instances according to conditions that you define
- Detects impaired EC2 instances and unhealthy applications, and replaces the instances without your intervention
- Provides several scaling options – Manual, scheduled, dynamic or on-demand, and predictive

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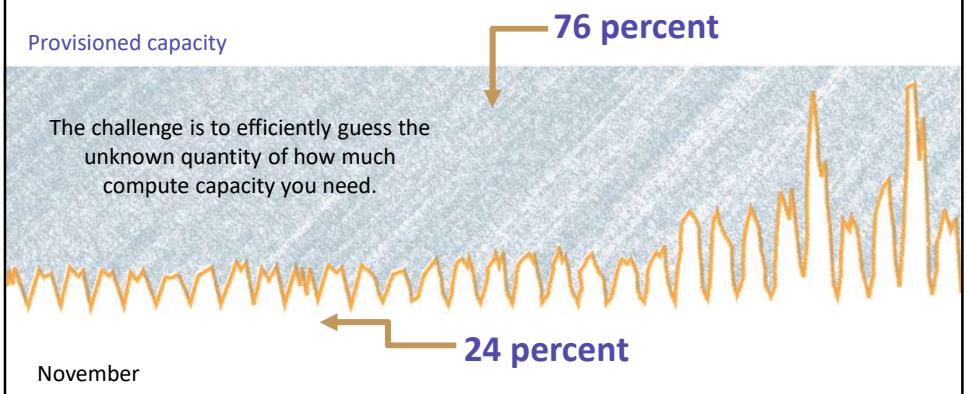
Typical weekly traffic at Amazon.com



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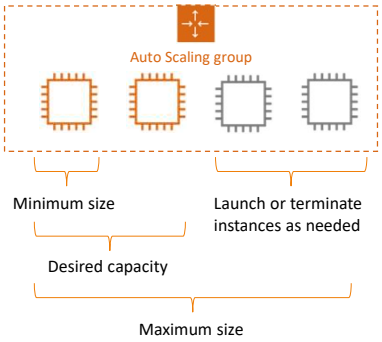
November traffic to Amazon.com



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Auto Scaling groups

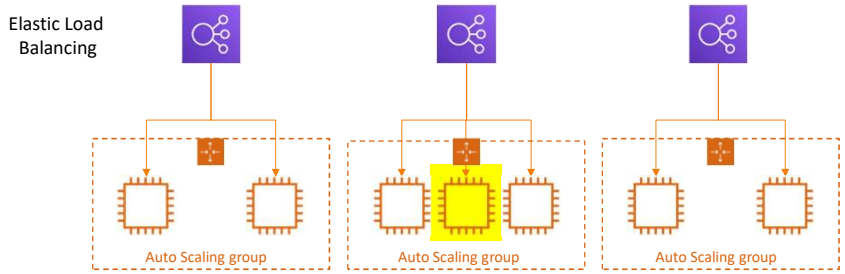


An **Auto Scaling group** is a collection of EC2 instances that are treated as a logical grouping for the purposes of automatic scaling and management.

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Scaling out versus scaling in



Base configuration

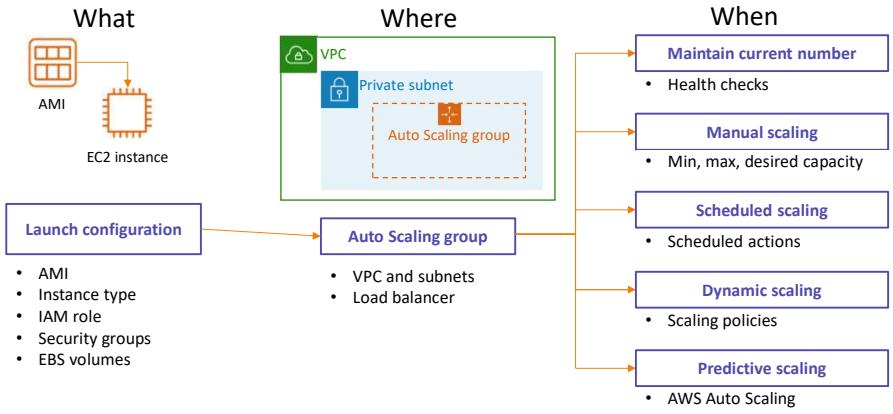
Scale out (launch instances)

Scale in (terminate instances)

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How Amazon EC2 Auto Scaling works



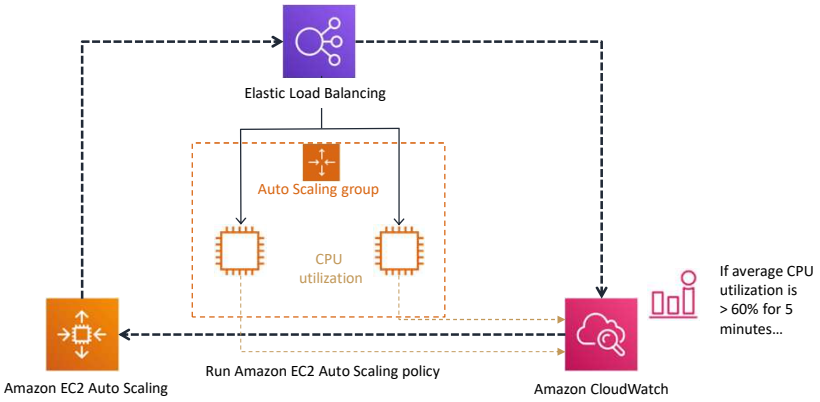
- AMI
- Instance type
- IAM role
- Security groups
- EBS volumes

- VPC and subnets
- Load balancer

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Implementing dynamic scaling



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AWS Auto Scaling

AWS Auto Scaling

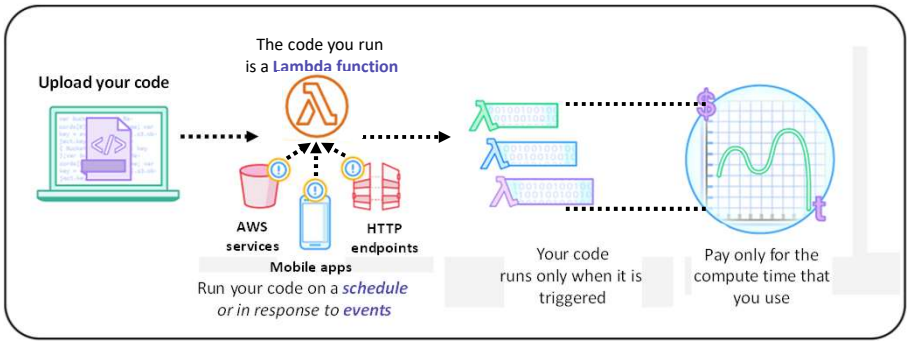
- Monitors your applications and automatically adjusts capacity to maintain steady, predictable performance at the lowest possible cost
- Provides a simple, powerful user interface that enables you to build scaling plans for resources, including –
  - Amazon EC2 instances and Spot Fleets
  - Amazon Elastic Container Service (Amazon ECS) Tasks
  - Amazon DynamoDB tables and indexes
  - Amazon Aurora Replicas

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AWS Lambda: Run code without servers

AWS Lambda is a **serverless** compute service.



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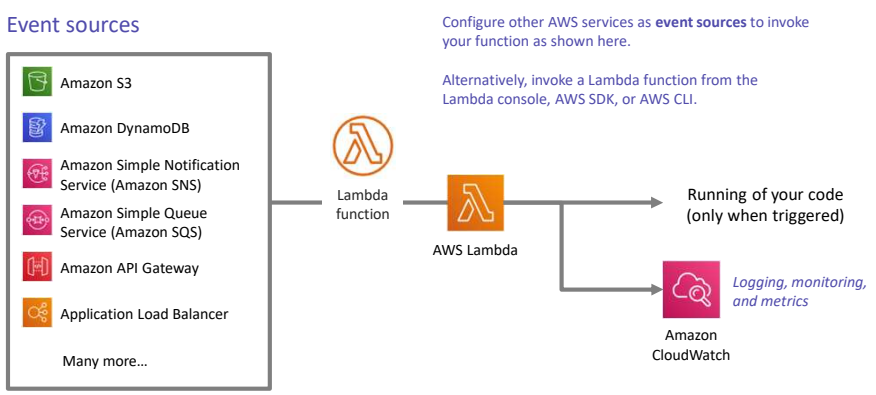
Benefits of Lambda

AWS Lambda

- It supports multiple programming languages
- Completely automated administration
- Built-in fault tolerance
- It supports the orchestration of multiple functions
- Pay-per-use pricing

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AWS Lambda event sources



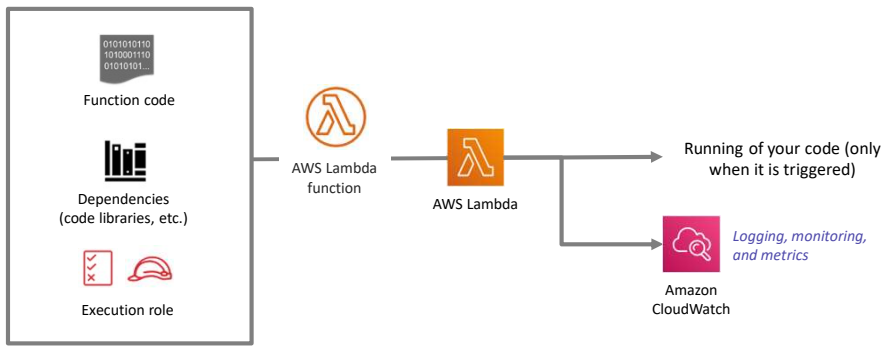
Configure other AWS services as **event sources** to invoke your function as shown here.

Alternatively, invoke a Lambda function from the Lambda console, AWS SDK, or AWS CLI.

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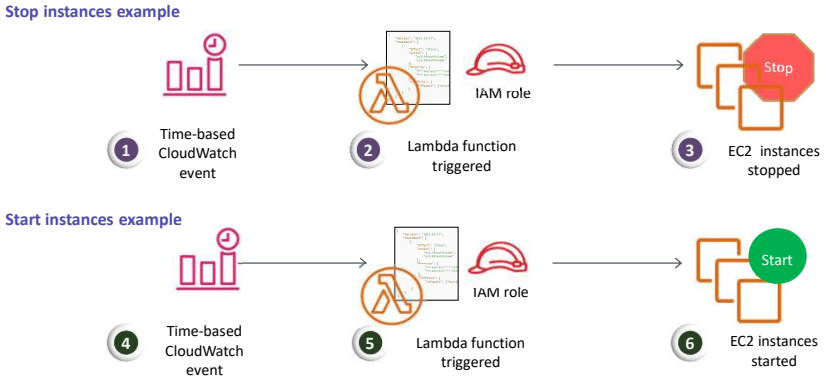
AWS Lambda function configuration



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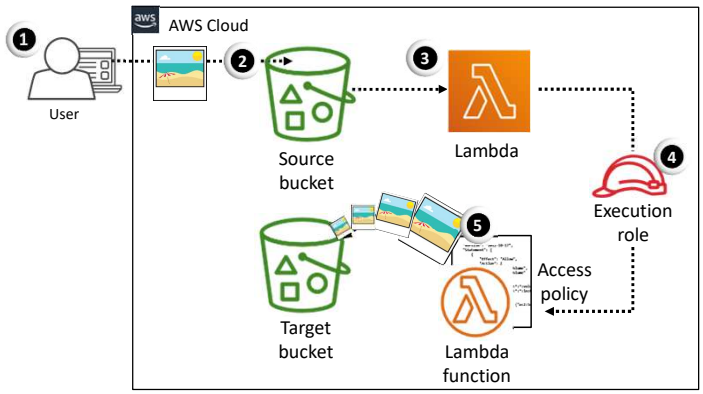
Schedule-based Lambda function example: Start and stop EC2 instances



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Event-based Lambda function example: Create thumbnail images



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AWS Lambda quotas

Soft limits per Region:

- Concurrent executions = 1,000
- Function and layer storage = 75 GB

Hard limits for individual functions:

- Maximum function memory allocation = 10,240 MB
- Function timeout = 15 minutes
- Deployment package size = 250 MB unzipped, including layers
- Container image code package size = 10 GB

Additional limits also exist. Details are in the AWS Lambda quotas documentation at https://docs.aws.amazon.com/lambda/latest/dg/gettingstarted-limits.html.

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AWS Elastic Beanstalk



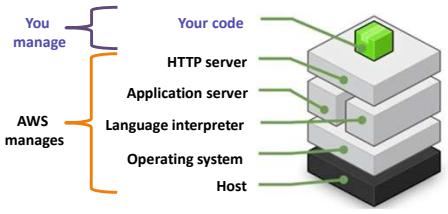
AWS Elastic Beanstalk

- An easy way to get **web applications** up and running
- A **managed service** that automatically handles –
  - Infrastructure provisioning and configuration
  - Deployment
  - Load balancing
  - Automatic scaling
  - Health monitoring
  - Analysis and debugging
  - Logging
- No additional charge for Elastic Beanstalk
  - Pay only for the underlying resources that are used

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AWS Elastic Beanstalk deployments

- It supports web applications written for common platforms
  - **Java**, **.NET**, **PHP**, **Node.js**, **Python**, **Ruby**, **Go**, and **Docker**
- You upload your code
  - Elastic Beanstalk automatically handles the deployment
  - Deploys on servers such as Apache, NGINX, Passenger, Puma, and Microsoft Internet Information Services (IIS)



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Benefits of Elastic Beanstalk



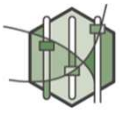
Fast and simple to start using



Developer productivity



Difficult to outgrow



Complete resource control

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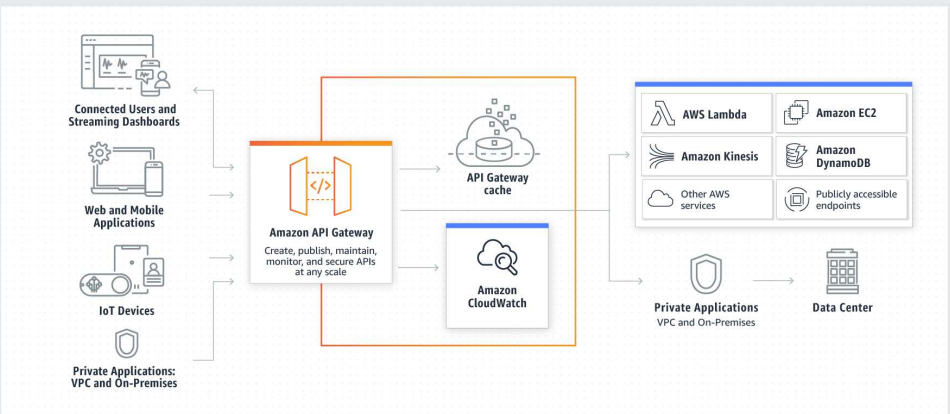
11. AWS Other Services

Ricardo Filipe, ISEC (ricardo.filipe@isec.pt)

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AWS API Gateway

Create, maintain, and secure APIs at any scale



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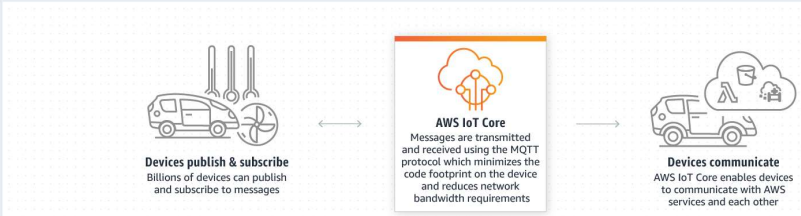
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AWS IOT

connect devices to the cloud

communication protocol, including MQTT, HTTPS, MQTT over WSS, and LoRaWAN.

We are going to use AWS IOT Core

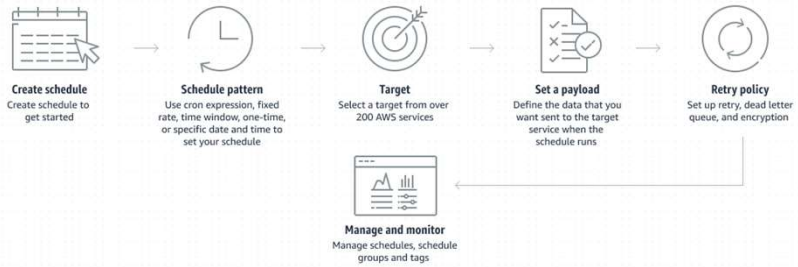


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Amazon EventBridge

Manage triggers to other AWS Services



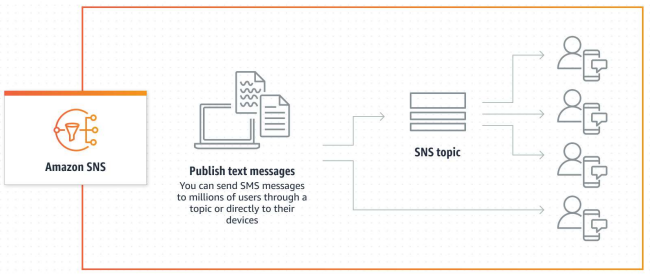
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Amazon Simple Notification Service

Amazon Simple Notification Service (Amazon SNS) sends notifications two ways,

- A2A (application to application) - event-driven
- A2P (application to person) – SMS, push notifications and email



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2. AWS Storage

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Storage

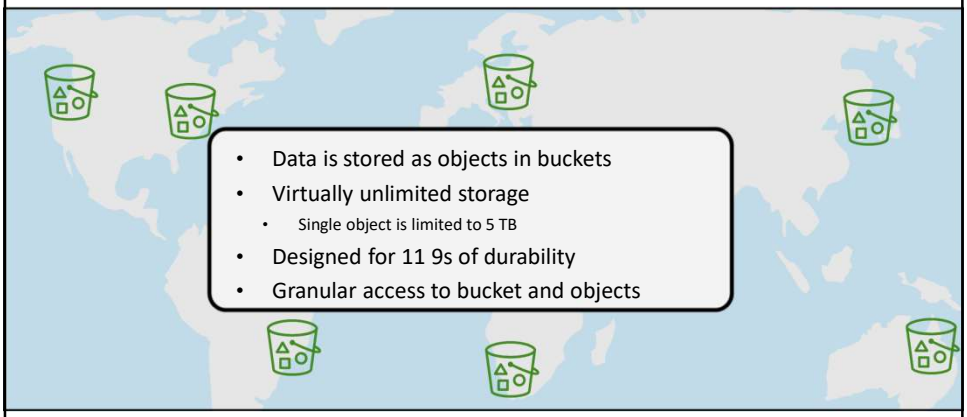


Amazon Simple Storage Service
(Amazon S3)

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Amazon S3 overview



- Data is stored as objects in buckets
- Virtually unlimited storage
  - Single object is limited to 5 TB
- Designed for 11 9s of durability
- Granular access to bucket and objects

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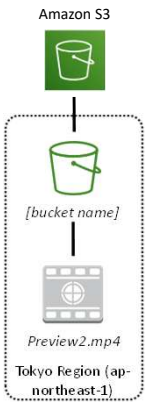
Amazon S3 storage classes

Amazon S3 offers a range of object-level storage classes that are designed for different use cases:

- Amazon S3 Standard
- Amazon S3 Intelligent-Tiering
- Amazon S3 Standard-Infrequent Access (Amazon S3 Standard-IA)
- Amazon S3 One Zone-Infrequent Access (Amazon S3 One Zone-IA)
- Amazon S3 Glacier
- Amazon S3 Glacier Deep Archive

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Amazon S3 bucket URLs (two styles)



To upload your data:

1. Create a **bucket** in an AWS Region.
2. Upload almost any number of **objects** to the bucket.

Bucket path-style URL endpoint:

https://s3.ap-northeast-1.amazonaws.com/bucket-name

Region code — Bucket name

Bucket virtual hosted-style URL endpoint:

https://bucket-name.s3-ap-northeast-1.amazonaws.com

Bucket name — Region code

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